

The Capacity of Trading Strategies *

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Abstract

Due to non-linear transaction costs, the financial performance of a trading strategy decreases with portfolio size. Using a dynamic trading model a la Garleanu and Pedersen (2013), we derive closed-form formulas for the performance-to-scale frontier reached by competitive traders endowed with a signal predicting stock returns. The decay with scale of the realized Sharpe ratio is slower for strategies that (1) trade more liquid stocks (2) are based on signals that do not fade away quickly and (3) have strong frictionless performance. We apply the framework to four well-known strategies. The capacity of strategies has *increased* in the 2000s compared to the 1990s due to increased liquidity. Because low volatility and past accounting profitability are persistent characteristics, strategies based on them are highly scalable, including in the mid-cap range. When traders underestimate the number of competitors trading a similar signal, their performance is strongly negatively impacted.

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1 Introduction

The empirical asset-pricing literature documents a wide array of variables that predict stock returns (Harvey et al., 2016). In various settings, these trading “signals” are shown to correlate with future returns in a statistically significant way. Such an analysis is typically done either by using Fama-MacBeth stock-level regressions or by computing Jensen’s alphas of sorted long-short portfolios, controlling for known risk factors. The interpretation of such findings is either that the Efficient Market Hypothesis fails, or that the benchmark asset pricing model is missing an important risk factor. Under both interpretations, statistical significance implicitly measures the size of the economic “anomaly”. The asset management industry can exploit these trading signals by offering investment strategies that can be viewed either as profitable arbitrage portfolios or as diversifying exposure to a new risk factor. However, even when the predictive power of the underlying signal is statistically significant, some of these investment strategies are difficult to implement at a reasonably large scale. The reason is that transaction costs increase more than linearly with the size of trading portfolios, which makes profitability shrink (Korajczyk and Sadka (2004), Frazzini et al. (2012)). This concern is especially acute for signals that work mostly with small caps – or equivalently for signals that work better in equal-weighted than value-weighted terms. It is also present when signals are not persistent enough, since they force investors to trade more often. Illiquidity and signal dynamics limit the extent to which asset managers can scale their investments up. Overall, in order to assess the economic significance of an asset-pricing anomaly, it is crucial to determine the amount that can be effectively invested on it, i.e. its capacity.

This paper offers a framework to compute and estimate the capacity of a trading strategy. The three key inputs are: (1) gross performance (the Sharpe ratio of the arbitrage portfolio absent transaction costs), (2) liquidity of underlying securities and (3) dynamics of the signal on which the strategy is based. Intuitively, strategies based on a highly predictive signal, trading more liquid securities, and on slow-moving signals, have greater capacity. To evaluate the relative importance of these three ingredients and measure capacity, we need to start from an optimal trading rule. To obtain it, we use an equilibrium version of Garleanu and Pedersen (2013). In this model, several investors are simultaneously trying to take advantage of the same return predictability. Each investor’s flow utility is equal to trading profits, minus a penalty for portfolio risk (scaled by risk aversion, as in the classical Markowitz model) and another penalty reflecting the cost of trading.

Investors crowd each other out via transaction costs. The main advantage of this model is that it rests on inter-temporal optimization, which turns out to be critical to study the effect of signal dynamics on capacity. Intuitively, investors optimally “slow down” trading to mitigate the impact of transaction costs and the cost of future unwinding.

We use this model to obtain a closed-form relationship between the trading scale of a strategy and its effective risk-adjusted performance. We define trading scale as the dollar volatility allocated to the strategy, aggregated over all investors. For a given trading scale, the model pins down the optimal portfolio at each point in time, and therefore its Sharpe ratio net of transaction costs. We call this performance-to-scale curve the “capacity” frontier: When a given number of traders collectively target a given P&L volatility, the capacity frontier gives the net of cost Sharpe ratio that they achieve.

We obtain closed form parametric equations describing the performance-to-scale frontier. These equations are easy to plot with a computer. Before applying the model to standard strategies, we build intuition by deriving a simple formula that is valid under reasonable approximations. This formula directly connects the scale of the portfolio (measured by the dollar volatility of profits, Vol) and the realized Sharpe ratio SR :

$$Vol = \frac{2SR}{\lambda\phi^2} \left[\left(\frac{SR^*}{SR} \right)^{2/3} - 1 \right]^2. \quad (1)$$

where SR^* is the “pure” Sharpe of the strategy, i.e. assuming no trading costs. λ measures the price impact of traded stocks. ϕ measures the speed at which the signal underlying the trading strategy fades away. Intuitively, a trader aiming for a larger realized Sharpe needs to reduce her exposure (Vol is a decreasing function of SR). Capacity has an elasticity of 1 with respect to λ and an elasticity of 2 with respect to signal mean-reversion ϕ . For an investor ready to accept a performance degradation by 30% ($SR = 0.7SR^*$), the portfolio scale that can be reached is simply given by $Vol = \frac{SR^*}{10\lambda\phi^2}$. We also provide a simple formula that shows how much of the gross P&L should be lost in transaction costs by an optimizing trader with access to capital. These amounts are large: Typically, an investor ready to accept a performance degradation by 30% vis-a-vis the “pure” Sharpe should lose as much as 20% of her P&L in transaction costs.

We then *calibrate* this model on four classical “fundamental” strategies and two different pools of US stocks (large caps, and mid caps). The four strategies are: book-to-market value of equity or

“value” (Fama and French, 2006), minus rolling volatility or “low-vol” (Ang et al., 2009), momentum in stock returns (Jegadeesh and Titman, 1993; Carhart, 1997), and operating cash flows to assets, or “quality” (Novy-Marx, 2013; Asness et al., 2014). We use US data to estimate, for each strategy, the persistence of the signal and the gross performance. For each pool of stocks, we estimate the price impact coefficient. We then inject these parameters in the model, implicitly assuming that the data-generating process (DGP) and the price impact function used in the model are correct. We find that under this calibration, value and momentum have relatively small capacity. This is due to the fact that these strategies – at least using the simple signals we deploy here – have low pure performance, and short-lived signals. In contrast, low-vol and quality have large capacities, both in the large-cap and in the mid-cap spaces. This is due to the fact that (1) liquidity has considerably increased since the 2000s, (2) these strategies exploit slow moving signals and (3) their pure performance is reasonably high.

We finally investigate the robustness of the model to various assumptions. First, we calibrate the loss made by an investor who would make a mistake about the number of participating traders. We find that this loss is asymmetric: It is large when traders underestimate the extent of competition. In this case, they trade too fast and too aggressively, leading to high transaction costs due to excess crowding. The loss is smaller when traders overestimate competition: this is because competition naturally makes investors trade too fast for their common good. Overestimating competition imposes some restraint, which is a beneficial effect. Second, we investigate the impact on the performance-to-scale frontier of an error in the data generating process. While our model assumes that the signal is an AR(1), we simulate signals that have predictive power but a different time-series structure, and estimate the frontier resulting from an investor trading under the AR(1) hypothesis. We find that for a class of “long memory signals” (AR(k) where $k > 1$, like “low-vol”) realized capacity does not really differ from expected one. Finally, in this robustness section, we estimate the loss that a static trader would make compared to an investor that trades dynamically. We find this loss to be very large. The dynamic model encourages investors to anticipate future unwinding of positions, which ends up being an important effect.

This paper relates to the recent literature that studies the impact of trading costs on the performance of various strategies. This literature departs from the traditional market microstructure literature in that it focuses on price impact, rather than fees and bid-ask spreads. Garleanu and Pedersen (2013) develop a framework that produces an optimal trading rule given a strategy and

costs of execution. We use their results in order to compute the capacity of an optimally traded signal. Novy-Marx and Velikov (2014) propose a new rule to avoid trading “too much” when implementing a series of well-known strategies. Their insight is that investors should not react too quickly to changes in signals, in order to avoid useless round-trips. They do not focus on the persistence of strategies, as we do and do not investigate price-impact increasing in trade size (e.g. quadratic transaction costs) but rather focus on linear transaction costs. They find large capacity for momentum size and value anomalies. Using a similar approach for transaction cost calibration and portfolio construction, Chen and Velikov (2018) shows that anomalies with positive net returns in sample deteriorate, leading to negative net returns post-publication. Korajczyk and Sadka (2004) study how price impact deteriorates returns in trading momentum; Using a calibration, they find that capacity of value-weighted momentum (\$ 2 Bn) is much higher than its equal-weighted capacity (\$200 Mil.). But their optimization is static which leads to significant underestimation of capacity, as we show in our last Section. Frazzini et al. (2012) analyze actual trade data with both executed and intended trades to infer trading costs, and then analyze the performance of well-known strategies once these trading costs are taken into account. Using these data, they find larger capacity for these strategies than existing papers. Compared to these studies, our paper uses the dynamic model of Garleanu and Pedersen (2013), which features an investor that takes into account the future implications of its current trading. This turns out to be particularly important for fundamental signals. The dynamic optimization allows us to trade efficiently even for large capacities, so that net-of-transaction cost performance does not become negative in the large investment range, in contrast to Korajczyk and Sadka (2004) and Frazzini et al. (2012).

This paper also indirectly relates to the “skeptical” literature that asks whether published market anomalies are really present in the data. McLean and Pontiff (2016) show that many anomalies disappear as soon as they are published on SSRN, thereby suggesting that arbitrage capital moves in quickly and makes anomalies go away. Harvey et al. (2016) provide a recent discussion of why most trading strategies uncovered by the literature are actually not statistically significant, because of data-snooping bias. Our paper focuses on another line of criticism, which is that anomalies may exist statistically, but that no investor with significant capacity can take advantage of them. We find that this is not the case for a class of slow moving signals. We also find that the increase in stock liquidity between the 1990s and the 2000s has been large enough to compensate for the reduction in the performance of these strategies. So, even if the alpha generated by some anomalies is lower

today that in the 1990s, the “size” of these anomalies is as large, sometimes even larger than before. Finally, our analysis strongly underscores the fact that “quality” (Novy-Marx, 2013; Asness et al., 2014) is a very large anomaly, by far the largest of the four that we document here, in the sense that it can be traded at very high scales while still delivering a high net-of-costs Sharpe ratio.

The next section lays out the dynamic trading framework which is an equilibrium version of Garleanu and Pedersen (2013). We derive in this section an explicit formula for the link between \$ volatility of a strategy (the amount invested) and its Sharpe ratio. We define the notion of capacity as the level of \$ volatility that is consistent with a given target Sharpe ratio. We then investigate the effect of signal persistence on volatility, and how this interact with liquidity. We use approximations and make approximations to build intuition and get functional forms. Section 3 describes the data. Section 4 calibrates the model. Section 5 investigates the robustness of the model. Section 6 concludes.

2 Framework

2.1 Set-Up

2.1.1 Data Generating Process

In this section, we discuss the return generating process, which is essentially the same as Garleanu and Pedersen (2013) (henceforth, GP). Let s_t be a signal that is used to forecast returns. s_t is assumed to be a vector whose dimension is equal to the number of securities traded. We assume that this signal can be described as an AR(1) process with persistence parameter ϕ :

$$\Delta s_{t+1} = -\phi s_t + \epsilon_{t+1} \quad (2)$$

where ϕ is a scalar and $\Delta s_{t+1} = s_{t+1} - s_t$. This assumption on the signal’s behavior is important to solve the dynamic programming problem. We explore its consequences numerically in Section 5.2.

The signal has some forecasting power over returns. To simplify exposition, we assume that the signal at t is the only variable forecasting returns at $t + 1$:

$$r_{t+1} = p_{t+1} - (1 + r_f)p_t = B s_t + u_{t+1}. \quad (3)$$

In equation (3) we omit exposure to predictable risk factors that price returns. This omission keeps exposition simple. r_f is the risk-free rate. B is a diagonal matrix of potentially stock-specific scaling factors. Another consequence of equation (3) is that past values of the signal (i.e. s_{t-k} for $k > 0$) do not add useful information to predict r_{t+1} , conditionally on s_t . Note also that we follow GP in defining r_t as share price change (adjusted for dividends and splits) rather than returns for tractability of the dynamic optimization problem.

This assumption about the data-generating process fits the GP framework. Their model is written in the spirit of APT: there is a set of K factors, and each security may have a different –but constant – loading on each of the factors. In our paper, we assume that there is one factor per security, and that each security has a loading of B on this factor. So our model corresponds to a special case of theirs, and their formulae apply.

Last, a critical assumption behind equation (3) is that the forecasting power of the signal does not change over time. One possible alternative model would be that the regression coefficient B is itself noisy, i.e. that $r_{t+1} = (B + \eta_t)s_t + u_{t+1}$. Such a model accommodates better the fluctuations in the performance of the strategy based on the signal s_t . Since the focus of this paper is on transaction costs, it is natural to start with the simple model in equation (3). Also, the GP model needs a constant variance-covariance matrix, so our analytical results do not hold under this alternative assumption.

2.1.2 Portfolio Optimization

We consider a market equilibrium with N traders independently trading on the same signal s_t . We denote $x_{k,t}$ the position of trader k at time t , and $x_{-k,t}$ the sum of the positions of all traders except trader k . In addition we denote x_t the sum of the positions of all traders, i.e., the sum of $x_{k,t}$ and $x_{-k,t}$.

Transaction costs are quadratic and reflect potential interactions between traders. Each trader faces a per share cost of $L_t = \frac{\lambda}{2}\Sigma(\Delta x_t)$. We assume transaction costs to be linear in the number of shares traded by all N traders. They are also proportional to Σ , where Σ is the variance-covariance matrix of the residual u_t . λ is a scalar reflecting the illiquidity of the pool of stocks being traded. Overall, the most expensive stocks to trade are the most volatile ones and the already most traded ones.

We assume that traders are symmetric.¹ Each trader's flow utility is a mean-variance criterion with an absolute risk aversion γN . As we will see, this notation ensures that, keeping γ constant, an increase in N does not lead to an increase in the aggregate amount of \$ invested in the signal. Hence, we can separately implement comparative statics in the amount of dollars invested on the signal (through γ) and comparative statics in the intensity of competition (through N). We return to this in Section 5.1.

Each trader k selects her portfolio by solving a dynamic problem, taking the action of others as given. Every period, her flow utility is equal a mean-variance criterion minus trading costs (which themselves depend on other traders' actions). Every period, the trader optimizes dynamically over positions $(x_{k,t})$, while taking into account expected returns, risk and trading costs and the discount rate δ :

$$\max_{(x_{k,t+s})} E_t \left\{ \sum_{s \geq 0} \frac{1}{(1+\delta)^s} \left[-(\Delta x'_{k,t+s}) L_{t+s} + \frac{1}{1+\delta} \left(x'_{k,t+s} r_{t+s+1} - \frac{N\gamma}{2} x'_{k,t+s} \Sigma_u x_{k,t+s} \right) \right] \right\} \quad (4)$$

where $L_t = \frac{\lambda}{2} \Sigma(\Delta x_t)$ and $\Delta x_t = x_t - x_{t-1}$.

The FOC leads to the law of motion for the optimal portfolio composition of trader k :

$$\gamma_k x_{k,t} = \gamma x_t^m - \frac{\lambda(1+\delta)}{2} (2\Delta x_{k,t} + \Delta x_{-k,t}) + \frac{\lambda}{2} (E_t \Delta x_{-k,t+1} + 2E_t \Delta x_{k,t+1}) \quad (5)$$

where x_t^m is the aggregate Markowitz portfolio $(\gamma \Sigma)^{-1} B s_t$. Note that, when there are no trading costs (i.e. when $\lambda = 0$), the Markowitz portfolio is the optimal allocation in this model.

At symmetric equilibrium, $x_{k,t} = \frac{x_t}{N}$, so that we can aggregate the laws of motion into the law of motion of the aggregate trader:

$$x_t = x_t^m - \frac{\lambda(1+\delta)}{2\gamma N} (N+1) \Delta x_t + \frac{\lambda}{2\gamma N} (N+1) E_t(\Delta x_{t+1}).$$

This is the same law of motion as in the single trader (i.e., monopolistic) case of GP, except that we obtain an additional multiplicative factor $\frac{N+1}{2N}$ on two terms of the right hand side of the equation. As in their paper, we guess the following law of motion for the optimal aggregate portfolio:

$$x_t = (1-\tau)x_{t-1} + \tau \kappa x_t^m = (1-\tau)x_{t-1} + \tau x_t^{aim} \quad (6)$$

¹We explore the case of asymmetric traders in Appendix A.

where τ and x^{aim} are, respectively, the “trading rate” and the “aimed portfolio”. Injecting this formula and solving for τ and κ leads to the following equations:

$$0 = \tau^2 + \tau \left(\delta + \frac{\gamma}{\lambda} \frac{2N}{(N+1)} \right) - \frac{\gamma}{\lambda} \frac{2N}{(N+1)} \quad (7)$$

$$\kappa = \frac{\gamma}{\gamma + \frac{(N+1)}{2N} \lambda \phi \tau} \quad (8)$$

Our solution differs from GP in that we are allowing for traders interaction in equilibrium. When $N = 1$, we are back to their model with a monopolistic trader. As in their paper, τ is increasing with liquidity (λ), but decreasing with size (measured here through $1/\gamma$, as less risk averse investors endogenously take on bigger exposures). This is intuitive: Liquidity and small size allow investors to trade faster. Portfolio leverage $\frac{\kappa}{\gamma} = \frac{1}{\gamma + \frac{2N}{N+1} \lambda \phi \tau}$ also has intuitive comparative statics properties. It increases with size, since larger investors are by design less risk averse. It decreases with illiquidity λ and signal mean-reversion ϕ . These are all properties of the GP model, but our version of the model also allows comparative statics in competition N . When competition increases, traders trade faster and aim higher. They trade “too aggressively” because they internalize less and less the effect of their own trading on aggregate transaction costs.

2.1.3 Calibration Challenges

When bringing the Garleanu-Pedersen (henceforth GP) model to the data, four caveats arise, that we discuss here. First, the model omits costs due to broker fees and financing fees. These costs are linear in the amount of \$ traded, and hence fees per \$ traded do not depend on the pool of stocks traded. They thus shift all of our results on returns by the same amount, but do not affect the comparisons across pools of stocks (for instance, large and mid caps).

Second, the GP model omits shorting fees: While these can vary across stocks, they do not substantially differ across the pools of stocks that we study. True, as mid-caps are more likely to be on special, shorting fees tend to be higher for smaller stocks (see for instance Stambaugh et al. (2012) and Dreschler and Dreschler (2014)). However, we can check that for sufficiently large amounts traded this effect is second order compared to price-impact concerns. We do this using proprietary data on stock-level shorting fees obtained from a large asset manager. These fees are quotes obtained from its brokers. Brokers typically split stocks between “General Collateral” (GC)

and “Hard to Borrow” (HTB or “on special”). In our data, most stocks belong to the GC category: For these stocks, the shorting fee is the same (the “GC rate”) and hovers around 10bp annually. HTB stocks are those for which the shorting demand is unusually high, or for which stock lenders are difficult to find. For these stocks, the quoted shorting fee can be several percent in annualized terms. In Figure 1, we report the difference in cost between GC and HTB stocks during the 2012-2014 period, for three stock categories ranked by market capitalization: 1-500, 500-1,000, and 1,000-1,500. Panel A reports the average excess rate charged for all HTB stocks, removing all stocks for which the rate is above 2%. With this 2% cap, the average cost of shorting for hard-to-borrow stocks is around 10bp, even for the smallest category of stocks. The right panel (Panel B) plots the fraction of *extremely* HTB stocks, i.e. stock for which the extra lending fee is above 2%. This fraction lies between 2% for the least liquid pool, and almost 0% for the most liquid pool. The bottom line is that (1) even illiquid stocks are cheap to short (Panel A), and (2) extremely HTB stocks are a tiny fraction of the stock universe (Panel B). On average over the entire universe, the data reveals that the gap in shorting costs between top 500 US stocks (ranked by market cap) vs. stocks that are in the (1000-2000) size range is less than 5bps annualized, which for a strategy with unlevered returns above 1% implies a Sharpe reduction of less than 5 percent.

Third, the GP model omits some rebalancing trades in the objective function by assuming that price changes as opposed to returns are normally distributed. When implementing the model, such rebalancing trades will however show-up as the variance-covariance matrix is evaluated using rolling windows. This will bias simulation results regarding the Sharpe deterioration in a conservative direction as our trades won’t be optimal vis-a-vis a more realistic dynamic structure of variance (see Collin-Dufresne et al. (2012)).

Fourth, our trading-cost model assumes no permanent price-impact (i.e. prices revert instantaneously after trading). It is possible to introduce slow reversal of price-impact in GP, but this comes at the cost of not having closed form solutions any more (a very attractive feature of the GP framework). Introducing such costs would lead us to be somewhat more conservative on capacity estimates as it would force traders to slow down their trading further. Brokmann et al. (2014) show that full reversal of prices post trading typically takes a few days. By contrast Frazzini et al. (2012) find in their trading data that 70% of price-impact is permanent.

2.2 Defining Capacity

2.2.1 The Sharpe-to-volatility Frontier

We now use equations (6)-(7)-(8) to derive an explicit formulation for the Sharpe-to-volatility frontier. These equations describe the dynamics of the position of the competing traders, which we use to compute the total \$ volatility of their positions, and the expected returns net of transaction costs. Then, for a given level of “total” risk aversion γ , we can compute the volatility and risk-adjusted and cost-adjusted Sharpe. The following proposition describes the formulae:

Proposition 1. *Assume that $B^2 E s_t' s_t \ll \Sigma$, i.e. stock returns predictability is low. Then, in steady state, for a given risk aversion γ , the aggregate portfolio of the N competing traders has \$ volatility and net-of-cost Sharpe ratios given by:*

$$\begin{aligned} Vol(\gamma) &= \frac{1}{(\gamma + \phi \lambda \tau \frac{N+1}{2N})} \left[\frac{\tau}{(2-\tau)} \frac{1 + (1-\phi)(1-\tau)}{1 - (1-\phi)(1-\tau)} \right]^{1/2} SR^* \\ SR(\gamma) &= \left[1 - \frac{\phi \lambda \tau}{(\gamma + \frac{N+1}{2N} \phi \lambda \tau)} \frac{1}{(2-\tau)} \right] \left[\frac{1 - (1-\tau)^2}{1 - (1-\tau)^2(1-\phi)^2} \right]^{1/2} SR^* \end{aligned}$$

where SR^* is the Markovitz Sharpe ($SR^* = E(s_t' B \Sigma^{-1} B s_t)^{1/2}$) and τ is function of γ given by equation (7).

Proof. See calculations in Appendix B □

The above parametric equations thus describe the Sharpe-to-volatility frontier: N traders with “collective” risk aversion γ and optimizing individually reach a risk-adjusted performance $SR(\gamma)$, and a volatility $Vol(\gamma)$.

In the parameter range we explore, the capacity frontier is quite intuitively a decreasing curve on the Sharpe-volatility space: Bigger \$ volatility exposure comes with lower performance net of transaction costs. This happens because the net-of-cost Sharpe ratio is an increasing function of γ , while \$ volatility is a decreasing one. These two comparative statics essentially happen because of the effect of γ on the aimed portfolio. Remember from equation (8) that the aimed portfolio is given by $\frac{\Sigma^{-1} s_t}{\gamma + \frac{N+1}{2N} \phi \lambda \tau}$. Leverage decreases in γ through two channels: (1) The direct γ term which is the standard Markowitz effect that leverage decreases with risk aversion and (2) the dynamic trading effect whereby risk-averse traders need to trade faster (risk-averse traders avoid deviating from the “pure” strategy as it is risky, hence τ is an increasing function of γ) and so cannot afford to

aim too close to the Markowitz portfolio. Because of these two channels, more risk averse investors tends to have a lower \$ volatility exposure, and since they trade less, a higher net of cost Sharpe ratio. Hence, on the Sharpe-volatility space, the parametric equations of Proposition 1 describe a decreasing curve, on which traders move to the right (lower sharpe, high \$ vol) as their risk aversion decreases (γ goes to zero).

The above parametric equations are easy to plot using a computer. They give us how much performance loss one can expect from increasing \$ volatility by a given amount. Plotting them on a graph requires knowledge of the parameters SR^* , λ , ϕ , N and δ . We will do this in Section 3. But before doing this we would like to explore the comparative statics properties of the capacity frontier. In next Section 2.2.2, we build intuition using valid approximations. In the following Section 2.3, we are directly using the parametric equation and evaluate comparative statics numerically.

2.2.2 Approximated Formulae

In this Section, we use two approximations to rewrite Sharpe-to-volatility frontier in a more easily interpretable way. The first approximation assumes γ small enough, so that the optimal trading rate τ is at most a few percentage points per day. The second approximation focuses on slow-moving signals. Both approximations are valid for the range of parameters and the types of signals we explore. They have the advantage of helping to build intuition about the comparative statics properties of the capacity frontier.

Let us first discuss the “large investment approximation”:

Assumption 1. Large Investments Approximation

We assume:

1. $\left(\frac{\gamma}{\lambda}\right)^{1/2} \ll 1$.
2. $\delta \ll \left(\frac{\gamma}{\lambda}\right)^{1/2}$

These two assumptions are easy to interpret. When they both hold, it is easy to show that $\tau \approx \sqrt{\frac{\gamma}{\lambda} \frac{2N}{N+1}}$. Hence, the first approximation means that the effective trading rate has to be small: At most a few percents of the portfolio can be traded every day. The second interpretation means that discounting issues (at the daily horizon, again) are negligible compared to the trading rate. In our applications, both assumptions are easily satisfied.

Further, we assume that the signal is reasonably slow:

Assumption 2. *Slow Signal Approximation*

We assume that $\phi \ll 1$.

This second approximation is valid for most classical strategies. This is because ϕ measures the speed of mean-reversion at the *daily* frequency. Even for a signal with a half-life of 10 days, we have that $\phi = 1 - .5^{1/10} = .067$. Hence, for all strategies using monthly signals, the “slow signal” approximation is valid. For the signals that we study in this paper, ϕ is in the 10^{-3} range. The daily mean reversion, which essentially sort stocks by their past day return (Khandani and Lo, 2008), would not qualify for this approximation, since there $\phi \approx 1$ as daily stock returns are close to being i.i.d.

Using the parametric equations of Proposition 1 and approximations 1 and 2, we obtain the following results:

Proposition 2. *Assume the “large investment” and the “slow signal” approximations hold and let the number of traders $N \rightarrow \infty$:*

- *The large investment approximation rewrites as a function of observable quantities: $\sqrt{\frac{2SR}{\lambda Vol}} \ll 1$ and $\sqrt{\frac{2SR}{\lambda Vol}} \gg \delta$.*
- *The targeted \$ volatility is an explicit function of the pure and targeted Sharpes, the signal persistence and the pool’s liquidity:*

$$Vol = \frac{2SR}{\lambda \phi^2} \left[\left(\frac{SR^*}{SR} \right)^{2/3} - 1 \right]^2$$

For a given targeted Sharpe, the \$ volatility has an elasticity of 2 w.r.t. signal persistence ϕ , and an elasticity of 1 w.r.t. to price impact λ .

- *The above expression makes the \$ volatility a decreasing function of the targeted Sharpe, i.e. the capacity frontier is decreasing.*
- *For a given \$ volatility, performance decreases faster with liquidity when the signal is less persistent:*

$$\frac{\partial^2 SR}{\partial \lambda \partial \phi} < 0$$

- Transactions costs are a simple fraction of the targeted sharpe of the strategy:

$$\frac{TC}{ER} = \frac{\phi}{\phi + \tau} = \left[1 - \left(\frac{SR}{SR^*} \right)^{2/3} \right]$$

where ER is the steady state expected \$ gross profit (i.e. net profit plus transaction costs) from the strategy.

Proof. See Appendix C. □

Proposition 2 clarifies economic intuitions behind the generic capacity frontier described in proposition 1. The first bullet point defines the range for which the “large investment approximation” is valid. In the following, we will assume $\delta \approx 8.10^{-5}$, which corresponds to an annualized discount rate of 2%. Assume a price impact parameter $\lambda \approx 10^{-5}$ (which correspond to our estimate of impact in the mid cap range, see Section 3). Assume further that we are targeting a Sharpe ratio of 0.5. Then, the first bullet point of Proposition 2 simply states that, as long as Vol is at least 100m\$, and below 40bn\$, the two parts of large investment approximation are easily satisfied.² Hence, in the region of strategies we will explore, this approximation is valid. Of course, our empirical analysis below uses the parametric equations of proposition 1, but this analysis helps clarify intuitions.

Using these two approximations, we can combine two equations of Proposition 1 into an explicit expression of the \$ volatility for a targeted sharpe ratio SR . This expression is shown in the second bullet point of the above proposition. Capacity increase with the “pure” Markowitz Sharpe SR^* , and decreases with transaction costs λ and signal mean-reversion ϕ . Somewhat intuitively, \$ volatility is quite sensitive the pure Sharpe. Assume for instance that the investor aims for an effective Sharpe of .5. Then, the term in $2SR[(SR^*/SR)^{2/3} - 1]^2$ will be worth .016 if the pure Sharpe is .6 (as for instance for the low vol strategy, see below), and .22 if the pure Sharpe is .9 (as for the cash-flow strategy, see below). Thus, our model suggests that, even if one assumes equal persistence and liquidity, the cash-flow strategy will have more than 10 times as much capacity as the low vol strategy, simply because its pure performance is better. Capacity is also quite sensitive to liquidity. Moving from the largest stocks to midcaps leads to an 8-fold increase in price impact. So, other

²If Vol = 100m\$, then

$$\sqrt{\frac{2SR}{\lambda Vol}} = .03 \ll 1$$

while if Vol = 40bn\$, then

$$\sqrt{\frac{2SR}{\lambda Vol}} = .002 \gg 8 \times 10^{-5}$$

things equal, capacity would be divided by 8. Last, capacity is highly sensitive to signal speed. Going from a half-life of 1 month ($\phi = .034$) to a half-life of 3 months ($\phi = .01$) would increase capacity by a factor $3.4^2 \approx 10$. Slower signals are easier to trade at larger capacities. Last, the second bullet point yields a simple rule of thumb to compute the capacity of a strategy, assuming a targeted 30% Sharpe loss. In this case, $SR/SR^* = .7$ and the dollar volatility is

$$Vol = \frac{SR^*}{10\lambda\phi^2} \quad (9)$$

which is simple, and easily delivers the right orders of magnitude for the class of strategies we explore in this paper, as we will see in Section 4.

The fourth bullet point shows that, for given \$ volatilities, the Sharpe reduction due to trading costs $\partial SR/\partial \lambda$ is bigger (more negative) when the signal is less persistent. This equation embodies an important effect, i.e. that “slow-moving” strategies are relatively more scalable when stocks are more liquid.

The fifth bullet point gives a useful intuition about the levels of transaction costs. The first part of the equality shows that how much expected profits are “burnt” through transaction costs as a simple function of targeted sharpe loss. For instance, a trader targeting a Sharpe loss of 30% would incur transaction costs that are $1 - .7^{2/3} = 21\%$ of expected profits. When the signal reverts more quickly, effective transaction costs are bigger. The formula also highlights that, when the trader trades more quickly, transaction costs are lower. This second effect comes from the fact that fast traders in the model are the ones that care about transaction costs (compared to risk). They prefer low capacity, and low trading costs. Note in passing that the first equality is valid even outside of the “slow signal” approximation (see proof). Going back to the example of daily mean-reversion discussed above, assuming $\phi \approx 1 \gg \tau$ we obtain that $TC/ER \approx 1$: Even if they are traded optimally, the profits of high-frequency strategies are almost entirely wiped out by transaction costs.

2.3 Comparative statics

In this section, we investigate the different comparative statics of the Sharpe-to-volatility frontier with respect to the different parameters of the model. In order to abstract from the approximations of the previous Section, we go back to the parametric equations of proposition 1. Figures 2, 3 and 4 show graphically the effects of ϕ , λ and N on the Sharpe-to-volatility frontier defined by the formula

in Proposition 1.

Figure 2 investigates the effect of liquidity. We consider a fictitious strategy whose frictionless Sharpe ratio is equal to 1, and whose persistence $\phi = 2 \cdot 10^{-3}$, which roughly corresponds to the typical persistence of our fundamental signals (see data Section below). For now, we set $N = 1$, i.e., monopolistic trading. We use 4 different values of λ , which correspond roughly to the median λ of mid-size caps (see data Section below for definitions and measurement) in 1991-1995 ($3 \cdot 10^{-4}$), 1996-2000 ($10 \cdot 10^{-5}$), 2001-2005 ($35 \cdot 10^{-6}$) and 2013-2017 ($26 \cdot 10^{-6}$). We see there that liquidity has a very big impact on the capacity of strategies. For instance, the Sharpe loss due to trading \$ 5bn could be roughly as large as 90% with the liquidity level of the early 1990s, it is only .5 with the liquidity level of the late 2000s. This small exercise suggests that the increase in liquidity witnessed in the 2000s –in particular in the mid-cap range– led to a considerable increase in the capacity of the fundamental strategies we are studying here. This will be confirmed by calibrations below.

Figure 3 highlights the effect of the speed of signal mean-reversion ϕ on the scale-performance frontier. Again, we consider a fictitious strategy whose frictionless Sharpe ratio is equal to 1, and a pool whose price impact $\lambda = 26 \cdot 10^{-6}$, which corresponds to the liquidity estimated in the mid pool over the last five years of our sample from 2013 to 2017 (again, see our data Section below). Again, we set $N = 1$ and now use 4 different values of ϕ , which correspond to the ϕ of book-to-market among large caps ($0.4 \cdot 10^{-3}$, the most persistent strategy), book-to-market among mid-caps ($0.7 \cdot 10^{-3}$), low vol among large caps ($1.5 \cdot 10^{-3}$) and low vol among mid-caps ($1.8 \cdot 10^{-3}$). ϕ has also a discernible impact on capacity. Assuming the target is a volatility of \$ 5bn, going from the least (low vol, mid caps) to the most persistent strategy (book-to-market large caps), the Sharpe degradation decreases roughly from 60% to only 10%. However, note that in Figure 3, we focus on a group of similarly slow strategies. More “transient signals” have little capacity. In particular, for momentum, we find ϕ to be around $6.9 \cdot 10^{-3}$, about four times bigger than the mid-cap low vol signal discussed here.

Finally, Figure 4 focuses on the effect of trader competition. As above, we consider a fictitious strategy whose frictionless Sharpe ratio is equal to 1 together with signal decay $\phi = 2 \cdot 10^{-3}$, and a pool whose price impact coefficient is $\lambda = 26 \cdot 10^{-6}$. We consider 4 different values of N : $N = 1$ which corresponds to the monopolistic case, duopoly $N = 2$, $N = 10$ and $N = 1000$ (virtually similar to the case $N = \infty$). As expected, the capacity frontier shifts “down” as N increases: When numerous, traders internalize less the impact of their trades everyone’s transaction costs. Everyone trades too fast which increases transaction costs. But more importantly, Figure 4 highlights that the capacity

frontier is not very much affected by competition. For instance, the Sharpe loss due to trading \$ 5bn is about 0.6 with 1,000 traders, versus only 0.5 in the monopolistic case. This suggests that capacity is much more affected by the amount of dollar that is put at work (i.e., γ in our configuration) than by the number of traders actually using the signal.

3 Data & Calibration

3.1 Data

To investigate returns gross of transaction costs, we follow the literature and use split- and dividend-adjusted monthly returns from CRSP and annual accounting variables from COMPUSTAT. Our period goes from August 1990 to June 2018. When we move to the optimal trading analysis, we use daily returns from CRSP. We believe that it is important to allow investors to trade at the daily frequency in order to account for the potential fast decay of a signal's predictive power.

We start with monthly data. From the CRSP universe of stocks, we extract two pools: “Large” and “Mid”. Every month, we sort stocks by market capitalization computed at the end of the previous month. “Large” is the set of the largest 500 stocks. “Mid” is the set of stocks ranking from 501 to 1500 in terms of this measure of size. Overall, the two pools change composition every month, but turnover is low. Every month, on average 1.2% of the stocks leave and enter the “large” pool. Monthly turnover is 2.4% in the “mid” pool.

Table 1, panel A, gathers the main descriptive statistics of our data. The average turnover (annual \$ volume divided by market capitalization) is similar in the large and mid pool (25% among large stocks, versus 31% among mid caps). But midcaps are much smaller than large caps and therefore offer much less room for capacity. The annual volume traded in all mid caps is about \$ 4.2tn annual, or only 20% of the large caps volume (\$ 17.7tn).

3.2 Definition of Anomalies Traded

We focus on four well-known stock-market anomalies. Signals are updated monthly. Let us denote t the month of trading.

- First, we compute a standard “*Value*” signal, which is the ratio of book value of equity to market value of equity (Fama and French, 2006). Book value of equity (item CEQ in COM-

PUSTAT) is taken from the most recent annual accounts corresponding to the fiscal year ended in month $t - 7$.³ Market value is computed as the end of December of the last calendar year.

- Second, we compute a “*Low Vol*” signal, which is equal to *minus* the volatility of daily returns computed using returns from month $t - 4$ to month $t - 1$. This signal uses the well-documented fact that low volatility stocks tend to perform well in the long run. Frazzini and Pedersen (2013) use the stock’s β as a measure of its riskiness. Ang et al. (2009) use a measure of idiosyncratic volatility closer in spirit to the one we use here. One interpretation for this anomaly is that these stocks provide “embedded leverage” to investors who are not allowed to borrow.
- Third, we compute a “*Momentum*” signal which is equal to cumulative return between $t - 12$ and $t - 2$ of the stock (Carhart, 1997).
- Finally, the fourth signal we look at is the “*Cash-Flows*” signal, which is equal to net operating cash flows (item OANCF in COMPUSTAT) normalized by total assets (item AT). These accounting items are taken from the last available annual accounts available 7 months before the current month. The economic intuition as to why this signal predicts returns is not fully understood: Novy-Marx (2013) hypothesizes that return on assets—which our cash-flow measure approximates—captures some measure of risk exposure of the firm; and therefore predicts returns. Bouchaud et al. (2019) propose a theory and tests based on investors’ systematic underreaction to news.

We then normalize each of these signals in the following manner. First, within each pool, we sort stocks by signal as available at the end of the previous month. We then calculate the rank of each stock, and normalize it so that it lies -0.5 and $+0.5$. In the case of momentum, we apply a simple dynamic risk-scaling of the signal, namely, we divide the normalized signal by the average volatility over the previous month of all stocks belonging to the trading pool: such dynamic scaling is needed to get a reasonably good sharpe ratio of the momentum strategy over this time-period (see e.g. Carhart, 1997; Daniel and Moskowitz, 2013) for the analysis of similar management techniques of momentum crashes). For the other signals, we do not apply such scaling.

³Our assumption is therefore that the information available today from COMPUSTAT becomes available to investors 6 full months after the end of the fiscal year.

3.3 Calibrating the Price Impact Parameter λ

We now calibrate the transaction cost parameter λ . We follow here Garleanu and Pedersen (2013): Engle et al. (2008) note that trades amounting to 1.59% of the daily volume in a stock have a price impact of about 0.10%. Using this approximation for each stock i , we can compute a liquidity parameter λ_i as the solution of:

$$1.59\% \times volume_i \times \frac{\lambda_i}{2} \times \sigma_i^2 = 0.1\%,$$

where σ_i is the daily volatility of stock i and $volume_i$ is its average dollar daily volume. This leaves us for each stock with the following formula:

$$\lambda_i = \frac{1}{8 \times volume_i \times \sigma_i^2}.$$

Using CRSP daily data, we compute for each year a stock-level λ_i . Separately for the “large” and the “mid” pools, we then define for each year the pool’s λ as the median of the liquidity parameters (λ_i) of stocks belonging to the pool.

Time-series changes in each pool’s λ are reported in Figure 5. We observe a strong decrease in transaction costs during 1990-2000 for both pools. Given the way we measure λ (proportional to the inverse of $Volume \times \sigma^2$), the sharp decrease in λ mostly comes from the observed increase in trading volume already documented for instance by Chordi et al. (2011) and Novy-Marx and Velikov (2014). Chordi et al. (2011) for instance document a fivefold increase in turnover between 1995 and 2009.

In our calibrations, we use the average values of λ in the last five full years of our sample, i.e. values from 2013-2017. Our goal is to examine the scalability of strategies in the context of contemporary liquidity conditions. This leads us to use $\lambda = 5 \cdot 10^{-6}$ in the large pool and $\lambda = 2.6 \cdot 10^{-5}$ in the mid pool. We report these numbers in Table 1.

3.4 Persistence Parameters ϕ

To estimate ϕ , for each signal $s_{i,t}$, we estimate the following regression on monthly data:

$$s_{i,t+1} = a + b \cdot s_{i,t} + \epsilon_{i,t}$$

via plain OLS. We estimate the above equation separately for each strategy, and for each pool of

stocks. This allows us to retrieve the daily persistence parameter $\phi = 1 - b^{1/20}$. In Table 1, we report the values of ϕ for each pool and each strategy. Looking at Table 1, several noteworthy features emerge. First, all these strategies are very persistent. The plain book-to-market signal is the most persistent of the four. For mid caps, the half-life of this signal ($-\log 2 / (250 * \log(1 - \phi))$) is about 3.5 years in the “mid” pool, versus 2 years for “low vol”. All signals appear to be slow-moving. Momentum is the least slow moving, with a half-life of about 5 months.

3.5 Scaling Matrix B

To estimate the signal scaling parameter B , separately for each strategy, we first run the following regressions on percent returns $r^{\%}$:

$$r_{i,t+1}^{\%} = B^{\%} . s_{i,t} + u_{i,t}^{\%} \quad (10)$$

using monthly data on percent returns and signal described above. The coefficient $B^{\%}$ is assumed to be the same across stocks for each strategy. It is estimated through OLS over the entire sample period 1990-2018. The simulation will therefore have some element of look ahead bias, but this is not critical here since, the focus of the study is the analysis of trading costs rather than the actual risk-adjusted performance of stock market anomalies.

The parameter we need to use is the diagonal scaling matrix B , which is the scaling factor of signal used to predict *dollar*, not percent returns. Hence, for each diagonal element i of the scaling matrix, multiplying the above equation (10) by p_{it} on both sides yields: $B = B^{\%} P_t$, where P_t is the diagonal matrix of stock prices. So B varies over time with P_t . Since our model assumes a constant scaling matrix B , this rescaling of the coefficient $B^{\%}$ creates a gap between the data and the model: We will use a time-varying matrix B in formulae that assume that B is fixed. So implicitly, we’ll assume our stationary model is good approximation of the non-stationary problem. We believe this approach is reasonable as we are assuming daily rebalancing.

3.6 Pure Markowitz Sharpe ratio SR^*

The parametric equations of the capacity frontier require us to compute the Markowitz Sharpe ratio of each strategy SR^* , which imposes an estimation of the variance-covariance matrix of dollar returns error terms $u_{i,t}$, Σ . This is in practice a difficult problem. We use a simple short-cut here.

We assume a simple one-factor structure for the error term in the *percent* returns generating process:

$$u_{i,t}^{\%} = \beta_i r_{M,t+1} + v_{i,t+1}$$

where $r_{M,t}$ is the percent market return and we assume that all shocks $v_{i,t}$ have the same variance σ^2 . We note σ_M^2 the variance of the common factor $r_{M,t}$ and β the vector of stock betas.

Given this structure, the variance-covariance matrix of the percent error terms is given by $\Sigma^{\%} = \sigma_M^2 \beta \beta' + \sigma^2 I$. Multiplying equation (10) on both sides, it is easy to see that the variance matrix of dollar return errors is $\Sigma = P_t \Sigma^{\%} P_t$, where P_t is the diagonal matrix of stock prices. Remember that our formulas assume Σ constant over time. So, as argued in the previous section, our mapping of data from % returns into the model implicitly assumes that the formulae from the stationary model are a good approximation for a situation where the scaling parameter and the variance covariance matrix change over time. We take comfort in the fact that they move slowly compared to our trading horizon.

Under this simple risk structure, it is easy to show (see Appendix F) that *dollar* positions of the Markowitz portfolio are given by the vector:

$$P_t x_t^m = \frac{B^{\%}}{\gamma \sigma^2} \left\{ s_t - \frac{\sigma_M^2}{\sigma^2} \frac{\sum_i \beta_i s_{it}}{1 + \frac{\sigma_M^2}{\sigma^2} (\sum_i \beta_i^2)} \beta \right\} \quad (11)$$

which can be easily interpreted. First, if stocks are volatile or the investor risk averse, the portfolio is less levered (the term in $\gamma \sigma^2$). Second, for a given signal, the Markowitz portfolio reduces exposure on high beta stocks in proportion to the covariance between the signal and stock betas, in particular if the market is more volatile. This is to reduce exposure to the risk factor. If the signal is orthogonal to stock betas, this second term is null. Third, the correlation between the betas and signals plays a critical role in the size of the beta and net dollar exposure of the portfolio. If signal and beta are uncorrelated, the Markowitz portfolio is market-neutral (it has a beta of zero).⁴ If beta and signal are negatively correlated (as in the low-vol anomaly), then the Markowitz portfolio has a negative beta and a dollar long bias. The intuition is that taking signal exposure mechanically creates short

⁴It is easy to see that the beta of the Markowitz portfolio is given by:

$$B^{\%} \frac{\beta' s_t}{\gamma \sigma^2} \left\{ 1 - \frac{\sigma_M^2 \beta' s_t}{\sigma^2 + \sigma_M^2 \sigma^2 (\beta' \beta)} \right\}.$$

It is equal to zero if β is uncorrelated with s in the cross-section (as the signal has a zero cross-sectional mean, $\beta' s_t = 0$).

exposure to the market, which is then optimally counterbalanced by adding weight to the long leg of the portfolio.

The parameters required to compute the Markowitz exposures are estimated every month on a rolling basis –thus we allow the variance-covariance matrix to move slowly over time, in contrast with the assumptions of our model. Using monthly returns data, we estimate all relevant parameters $(\sigma_M^2, \sigma^2, \beta)$ based on 24 months rolling windows: Every month t , for every stock i , we compute the volatility of monthly returns over the past 24 months. σ^2 is then estimated as the cross-sectional average over the pool. We also compute the univariate $\beta_{i,t}$ as the univariate beta of the stock's return with the market returns –net of the risk-free rate– over the past 24 months. Finally, we compute $\sigma_{M,t}^2$ as the volatility of monthly market returns (adjusted for the risk-free rate), over the past 24 months. As explained in Section 3.2, signals are also re-estimated on the first day of each month, so that the portfolio exposures are also reevaluated once a month.

For each of our 4 strategies, we then compute the realized monthly return of the Markowitz portfolio over the period we study. The return corresponds to the monthly realized \$ PNL of the Markowitz portfolio per \$ of long exposure. Formally, the monthly return is computed as:

$$R_t^m = \frac{(P_t x_t^m)' r_t^{\%}}{(x_t^m)' e_t^m}$$

where e_t^m is a vector of dummy variables equal to 1 if stock i is on the long leg ($x_{it} > 0$), and zero else. The denominator is thus the long \$ exposure of the portfolio, while the numerator is the \$ PNL of the strategy.

In Figure 6, we report the cumulative return of the Markowitz portfolio for each strategy and in each size pool. As discussed in the introduction, our 4 strategies tend to perform better on smaller capitalizations. We also report the realized annualized Sharpe ratios of these strategies in Table 1. Besides the Markowitz Sharpe, we also report the “raw” Sharpe, i.e. the long-short portfolio where each stock is simply weighted by the signal. Because the signal is uniform on $[-.5, .5]$, the portfolio is dollar-neutral and its return during period $t + 1$ is $P_t^{-1} s_t' r_{t+1}$. Looking at the Table, we observe three salient features. First, the raw Sharpe of book-to-market and low vol are extremely low, even slightly negative for large caps. Secondly, for all strategies, the performance is similar or higher in mid caps than in large caps. Thirdly, the Markowitz portfolios have much better Sharpe ratios than the raw portfolios: the hedging provided by mean-variance optimization does improve performance

a lot, from .08 to .7 in the case of low-vol in the mid cap space. This is all the more striking because our simple method puts quite a lot of structure on the variance-covariance matrix of returns –in particular, it assumes only one factor, as well as homoskedastic returns.

4 Calibrating the Sharpe-to-volatility frontier

In this Section, we use the parametric equations of proposition 1 and the calibrated parameters described in the previous Section to calculate the capacity frontier of the four strategies. We show two different presentations of our results: The first one is in Table 2, where we compute the \$ volatility attained by an investor aiming for a net-of-cost Sharpe ratio of .3. The second set of results consists of the capacity frontiers themselves (hence reporting the \$ volatility attained for different levels of Sharpe) and is shown graphically in Figure 7.

Let us start with the \$ volatility attained with a Sharpe of .3. We show two different values: the “competitive” calibration (assuming $N = 1000$) and the “monopolistic” calibration ($N = 1$). We then compute the \$ volatility as follows. For each strategy, each pool of stocks, and for two separate periods⁵ 1991-2001 and 2002-2017, we make γ vary between 0 and 10^{-8} and select the value leading to a realized Sharpe ratio as close to .3 as possible, using the formula in Proposition 1. For these values of γ we then compute the corresponding \$ volatility. Of course, when the hedged Markowitz Sharpe is below .3, we set the capacity to 0, since it is not feasible to reach the minimum Sharpe of .3.

We report the calibrated parameters as well as our measure of capacity in Table 2. Several salient facts can be noticed. First, two groups of strategies emerge: Market-to-book and momentum have low capacity, while low-vol and cash-flows have large capacities. The low capacity of momentum arises from the fact that the signal mean-reverts more quickly than the other strategies (with a ϕ parameter about 6 times higher than the three others). For market-to-book, the signal is much slower, but the pure Sharpe is low, which is why capacity is low, or even non-existent in the recent period. A possibility is that a combination of value and momentum à la Asness et al. (2013) would have more capacity in the U.S. equity space. In contrast to these popular anomalies, low-vol and cash-flows have large capacities because pure Sharpe ratios are high but also because the signals are very slow. In the recent period (2002-2017), capacities are of the order of \$50bn in the large cap

⁵We do this because, as we saw in Section 3.3, liquidity increased markedly in the 2000s, in particular for midcaps.

segment, and about \$13bn in the mid-cap segment. Remember that these numbers correspond to \$ volatilities, not assets under management. Assuming funds aim for about 10% of annual volatility, capacities in terms of AUMs would mechanically be about \$500bn in the large pool and \$130bn in the smaller pool. This is quite large. The second fact apparent in Table 2 is that the increase in liquidity appears to have had a big impact on the predicted capacities of strategies. This is true both in the mid-cap and large-cap ranges. For instance, in the large-cap pool, λ has nearly been reduced by a factor of 20. This has led to an increase in the capacity of “cash-flows” by a factor of 26, from \$1.6bn to \$43bn. Note that the two orders of magnitude should be similar given our approximated formula (9), where the elasticity of capacity to λ is one. The effect of increasing liquidity on capacity is even more salient in the mid-cap pool. In the 1990s, low-vol could only be traded at small capacity (\$160m according to our calibration). That number increased by a factor 100, in part due to improved performance but also because λ shrunk by a factor 6 in the 2000s. The capacity of cash-flows only increased by a factor 2 as the positive effect of more liquidity was counteracted by a reduction in the pure performance of the strategy in this pool. A third fact is the relatively modest, yet economically significant effect of competition. Going from monopolistic trading to nearly perfect competition typically reduces capacity by about 15%. This order of magnitude is roughly constant across strategies and pools.

We also report the full capacity frontiers in Figure 7, where we set $N = 1$ to clarify presentation. These frontiers are obtained by varying risk-aversion γ in the parametric equations. Since our calibrated parameters might be noisily estimated, we want to explore how sensitive our frontier estimate is to this noise. So we estimate standard errors for these frontiers via a simple bootstrap procedure. Specifically, over our whole sample, we first calibrate the model’s parameters as explained above and we draw, in blue in Figure 7, the corresponding Sharpe-to-volatility frontier for each anomaly in each stock pool, using the formulae in Proposition 1. As the frictionless Sharpe is negative for the book-to-market anomaly in the large pool, we do not draw any frontier for that configuration. We then obtain confidence intervals by block-bootstrap simulations. In each of these simulations, we draw a set of stock identifiers and then retrieve the history of each of these identifiers. We draw the same number of stocks as in the initial pool. We then re-estimate the model parameters on this fictitious sample, and use them to construct the Sharpe-to-volatility frontier for this draw. We then repeat this procedure 100 times. The confidence intervals, drawn as dashed lines in Figure 7, correspond, for a given configuration, to the top and bottom 5% Sharpe ratio obtained for each

level of volatility.

The set of possible Sharpe-to-volatility frontiers vary quite a bit across signals. Interestingly, we see that a realized Sharpe of say .2 is typically not outside these simulated 95% confidence intervals for “large” levels of volatility for low vol and cash-flow, meaning that realizing a track-record with such a Sharpe ratio does not seem unlikely even with a considerable amount of money invested. Clearly, the low persistence of the momentum signal makes the corresponding Sharpe-to-volatility frontiers decline quickly toward zero, even for somehow low levels of volatility. Finally, it is worth mentioning that the high variance of the frictionless Sharpe for the book-to-market in the mid pool highlights that the performance of this anomaly highly depends on the pool of stocks drawn in the simulation. As in most cases, the uncertainty about capacity largely stems from the uncertainty about the pure performance.

5 Robustness

To further explore the practical relevance of our framework, we explore how robust it is to various types of noise sources. First, we explore the effect of “crowding”: We assume that the model holds but that agents only have noisy evaluation of the actual number of competitors (N), and we ask how this impacts the performance of the trading strategy. Second, we consider the impact of “model noise”: We assume that the true data generating process is different from that assumed in the model, and ask how a trader using the model to trade would perform. Third, we examine how needed the dynamic optimization approach chosen in the paper is, by comparing its outcomes to those of a myopic trader that would just use static optimization.

5.1 Crowding and sensitivity to competition

In this subsection, we run the following thought experiment: The traders believe that the true value of N differs from its actual value. The result of these beliefs is that the traders do not trade optimally, as they either over-estimate or under-estimate the total flows invested in similar trades as their own. For instance, if the traders underestimate the number of agents, they will “aim too high”, i.e. they will trade too aggressively on the signal. This leads to high transaction costs due to excess crowding. Our model will help us quantitatively evaluate the impact of this mistake.

While N is the actual number of traders in the economy, we assume they believe that this number

is $\hat{N}$. Further, assume that the risk-aversion coefficient of each investor is symmetric and equal to γN . This is just a normalization since we will not perform comparative statics in N (only in $\hat{N}$). Otherwise, the optimization problem is the same as in (4) and leads to the same law of motion for trader k :

$$\gamma N x_{k,t} = \Sigma^{-1} B s_t - \frac{\lambda(1+\delta)}{2} (2\Delta x_{k,t} + \Delta x_{-k,t}) + \frac{\lambda}{2} (E_t \Delta x_{-k,t+1} + 2E_t \Delta x_{k,t+1}) \quad (12)$$

where $x_{-k,t}$ is the trader's belief about the aggregate trading of all competitors. Since the trader knows the equilibrium is symmetric, she believes this number to be given by $(\hat{N}-1)x_{k,t}$. Replacing in the above trading rule and aggregating over all N traders leads to the following motion for aggregate trading $x_t = N x_{k,t}$:

$$x_t = x_t^m - \frac{\lambda(1+\delta)}{\gamma} \frac{\hat{N}+1}{2N} \Delta x_t + \frac{\lambda}{\gamma} \frac{\hat{N}+1}{2N} E_t(\Delta x_{t+1}).$$

where $x_t^m = (\gamma \Sigma)^{-1} B s_t$ is the Markowitz portfolio. This expression is similar to the earlier law of motion. Therefore, we can reuse the algebra of Appendix B after replacing $(N+1)/N$ by $(\hat{N}+1)/N$. This leads to the following proposition:

Proposition 3. *Assume there are N traders, each of them believing they are $\hat{N}$. Then, the objective capacity frontier of the strategy writes:*

$$\begin{aligned} Vol(\gamma) &= \frac{1}{\left(\gamma + \phi \lambda \tau \frac{\hat{N}+1}{2N}\right)} \left[\frac{\tau}{(2-\tau)} \frac{1 + (1-\phi)(1-\tau)}{1 - (1-\phi)(1-\tau)} \right]^{1/2} SR^* \\ SR(\gamma) &= \left[1 - \frac{\phi \lambda \tau}{\left(\gamma + \frac{\hat{N}+1}{2N} \phi \lambda \tau\right)} \frac{1}{(2-\tau)} \right] \left[\frac{1 - (1-\tau)^2}{1 - (1-\tau)^2(1-\phi)^2} \right]^{1/2} SR^* \end{aligned}$$

where SR^* is the Markowitz Sharpe ($SR^* = E(s'_t B \Sigma^{-1} B s_t)^{1/2}$) and τ is function of γ given by:

$$0 = \tau^2 + \tau \left(\delta + \frac{\gamma}{\lambda} \frac{2N}{\hat{N}+1} \right) - \frac{\gamma}{\lambda} \frac{2N}{\hat{N}+1}.$$

We show the effect of $\hat{N} \neq N$ in Figure 8. The latter explores the effect of a mistake about N . In this first Figure, we assume that $SR^* = 1$, $\phi = .002$, $\lambda = 2.6 \cdot 10^{-5}$ (the price impact of midcaps). There are $N = 100$ traders operating on this market, but investors believe that there are $\hat{N}$ of them. The thick black line corresponds to the capacity frontier when traders are realistic: $\hat{N} = N$. We then

experiment with four values of $\hat{N}$: 20, 50, 200 and 500. In most cases, mistakes hurt performance net of transaction costs. This is because a mistake about aggregate trading by competitors leads investors to trade at the wrong speed, and to fail to aim at the right height. In some cases however (as in the case where $\hat{N} = 200$), this may improve capacity because overestimating the competition leads to some “optimal restraint”. In the realistic equilibrium, investors trade too fast because they fail to internalize the fact that they hurt each others’ transaction costs, as in the Cournot model of monopolistic competition. At the margin, making traders overestimating the number of participants forces them to trade less aggressively, which increases collective capacity: It indirectly makes them internalize their negative externality on the performance of others. Once however the overestimation is too large, investors slow trading more than what is collectively optimal, deteriorating portfolio performance, which decreases capacity. By contrast, under-estimating the number of competitors is always bad for capacity: Realized Sharpe ratios are below those expected by the traders, as they under-estimate the amount traded by their peers simultaneously. It is one way to think about “crowded trades”.

5.2 Model robustness to alternative signal process

In this section, we quantitatively evaluate the extent to which our assumption that the signal follows an AR(1) is important or not.

We run the following thought experiment: Imagine that the signal were to follow another process, let the trader evaluate an AR(1) on the data and trade according to the model. We then ask whether the realized Sharpe massively differs from that predicted by our trading model. We answer this question by performing Monte-Carlo simulations. We simulate data where the signal allows to forecast returns as in equation (3), but the signal comes from a non-AR(1) generating process. We then assume that the trader fits an AR(1) model on the signal data, and trades according to our model. We compute realized Sharpe and \$ volatility for this process, and compare them with the Sharpe and \$ volatility that would emerge if the trader’s model was correct (i.e. if the signal really was an AR(1)). These values are directly taken from the closed-form equations in Proposition (1).

More precisely, we consider auto-regressive processes with more lags:

$$s_{t+1} = \sum_{i=1}^L \frac{(1-\phi)}{L} s_{t-i} + \epsilon_{t+1}.$$

Our baseline model assumes $L = 1$. We simulate such process for $L = 2, 4, 6$. We do so on a panel of 1,000 stocks with same liquidity and time length as our mid-cap pool. We generate return data using the signal above and assuming that the predictive power of the signal is that of the cash-flows signal in real data and idiosyncratic volatility matches real data. For each of the dataset of signals and returns that is generated, we compute the parameter $\hat{\phi}$ that results from estimating in those synthetic data an AR(1) : $s_{t+1} = (1 - \phi)s_{t-i} + \epsilon_{t+1}$. Then, we trade according to the model's optimal rule and compute performance net of transaction costs for various target volatilities. We perform this for 30 synthetic (signal-return) draws. Finally, we report in Table 3 the average Sharpe ratio decay (i.e. SR/SR^*) that is reached at various volatility scales, and compare them to the prediction of the model given the persistence parameter $\hat{\phi}$ estimated from the data.

We find that the realized Sharpe ratios are quite similar to prediction, even for high \$ volatility. In fact, the realized Sharpe might even be higher than anticipated (for the AR(6), vol=15 configuration). The reason is that at low frequency, the signal is actually more persistent than what the estimation of a monthly AR(1) suggests, therefore trading costs are smaller than expected. This result does not depend on all coefficients in the AR(L) being equal.

All in all, this robustness exercise suggests that our modeling approach remains valid for signals that are AR(L) with positive coefficients. For such signals, the AR(1) estimation does not lead to trading realizations that are far from expectations. This is consistent with the idea that an AR(1) is not a bad approximation of more complex processes with short time-series (Fuster et al., 2010).

5.3 Myopic Trading

This final section shows that the dynamic setting of this paper cannot be adequately approximated by a myopic optimization. The dynamic apparatus is critical for the slow-moving signals that we focus on here. As we will see, a static approximation leads to very suboptimal trading.

To show this, we consider a “myopic” alternative to the model, where the investor does not take into account the future trading costs that his current trading decision would entail. This is, in some sense, the framework used in most existing studies of capacity (Frazzini et al. (2012), Korajczyk and Sadka (2004)): Traders optimize the expected risk-adjusted performance, taking into account the instantaneous impact of their trades. They do not, however, take into account the effect of current trading on future profits (if the signal persists) or future trading costs (if the signal mean-reverts).

To remain consistent with the paper's framework and previous notation, we model the myopic

optimization problem as:

$$\max_{x_{k,t}} E_t \left[-\frac{\lambda}{2} (\Delta x'_{k,t}) \Sigma (\Delta x_t) + \frac{1}{1+\delta} \left(x'_{k,t} r_{t+1} - \frac{\gamma}{2} x'_{k,t} \Sigma_u x_{k,t} \right) \right] \quad (13)$$

for each period t and taking $x_{k,t-1}$ as given. This linear-quadratic static optimization is easy to solve. Aggregating the optimal trading rule over all N traders, we obtain:

$$x_t = (1 - \pi) x_{t-1} + \pi x_t^M \quad (14)$$

where $x_t^m = (\gamma \Sigma)^{-1} s_t$ is the Markowitz portfolio and $\pi = \frac{\gamma}{\gamma + \lambda(1+\delta) \frac{N+1}{2N}}$ is the trading speed of the myopic agent.

To fix ideas, assume that the large investment approximation holds: $(\gamma/\lambda) \ll 1$ and $(\gamma/\lambda) \gg \delta$. Under this approximation, the trading speed becomes $\pi \approx \frac{\gamma}{\lambda} \frac{2N}{N+1} \ll 1$. Conditionally on this approximation, it is easy to see that the difference between the myopic and the dynamic model is twofold. On the one hand, the myopic investor trades too slowly. The dynamically optimal trading rate is $\tau = \left(\frac{\gamma}{\lambda} \frac{2N}{N+1} \right)^{1/2}$ while the myopic one is $\pi = \frac{\gamma}{\lambda} \frac{2N}{N+1}$. Second, the myopic investor aims “too high”, since he goes towards the Markowitz portfolio, which is more levered than the “aimed” portfolio of the GP model. The difference between the two targets is larger when mean-reversion ϕ is bigger, and when risk-aversion γ is bigger: In both cases, the myopic trader makes bigger mistakes because he does not take into account the future cost of exiting the position. Note that the difference also increases with N .

In order to see more clearly the effect on capacity and implement some simple quantification, we explicitly formulate the Sharpe-to-volatility frontier of the myopic trader in the following Proposition:

Proposition 4. *Assume the “large investment” and “slow signal” approximations hold, the Sharpe-to-volatility frontier realized by a myopic trader writes as:*

$$Vol = \frac{2N}{N+1} \frac{SR}{\lambda \phi} \left[\left(\frac{SR^*}{SR} \right)^2 - 1 \right]$$

Proof. See Appendix E. □

Let us consider trader aiming for a Sharpe of SR and trading a signal of mean-reversion ϕ . Let Vol_{Dyn} be the \$ volatility reached if he trades dynamically, and Vol_{Myo} the \$ volatility if he trades

myopically. In the large investment and slow signal approximations, the ratio of the two volatilities is given by the following expression as the number of traders $N \rightarrow \infty$:

$$\frac{\text{Vol}_{Dyn}}{\text{Vol}_{Myo}} = \frac{1}{\phi} \cdot \frac{[(SR^*/SR)^{2/3} - 1]^2}{(SR^*/SR)^2 - 1}$$

The myopic capacity frontier is much smaller than the dynamically traded frontier. This comes from the fact that, for our strategies $\phi \approx .001$ (see data Section). When $\frac{SR^*}{SR} \approx 1.1$ (the trader aims for a small reduction in performance), then $\frac{\text{Vol}_{Dyn}}{\text{Vol}_{Myo}} \approx 20$. When the trader is less ambitious in terms of Sharpe ratio, say, $\frac{SR^*}{SR} \approx 2$, then $\frac{\text{Vol}_{Dyn}}{\text{Vol}_{Myo}} \approx 100$. So in the range of parameters that we explore, not optimizing dynamically reduces capacity by a factor of 20 to 100. Of course, for shorter-lived signals the difference is much smaller.

6 Conclusion

Anomalies are "abnormal" only if substantial dollar amounts can be profitably put at work to arbitrage them. This paper explores the deterioration of the Sharpe ratio of a trading signal when the dollar scale of the traded portfolio increases. We show that, using dynamic optimization under quadratic transaction costs, yields simple formula for \$ volatility / Sharpe efficiency frontier, which we label the capacity frontier. When signals are persistent enough, arbitrageurs can put large amounts of money at work by trading more slowly.

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Table 1: Large vs Mid Caps

	Large	Mid
<i>Panel A: Sample Statistics</i>		
Total Capitalization (bn)	10163	1536
Total Volume (bn)	17770	4213
Average Turnover (%)	25	31
λ ($\times 10^{-6}$)	5	26
<i>Panel B: Strategy-level Statistics</i>		
Book-to-market		
Raw (unhedged) Sharpe	-.12	.01
Markowitz Sharpe	-.14	.06
ϕ (persistence $\times 10^{-3}$)	.5	.8
Low vol		
Raw (unhedged) Sharpe	-.03	.08
Markowitz Sharpe	.52	.7
ϕ (persistence $\times 10^{-3}$)	1.4	1.4
Momentum		
Raw (unhedged) Sharpe	.32	.27
Markowitz Sharpe	.39	.49
ϕ (persistence $\times 10^{-3}$)	6.8	6.8
Cash-Flows		
Raw (unhedged) Sharpe	.47	.7
Markowitz Sharpe	.52	.92
ϕ (persistence $\times 10^{-3}$)	.8	1.2

Note: This table reports summary statistics on the pools of “Large” and “Mid” caps. Every month, stocks are sorted by stock market capitalization. The largest 500 ones belong to the “Large” pool. Stocks ranking between 501 and 1500 belong to the “mid” pool. Panel A reports summary statistics for each pool: Aggregate volume in \$bn; aggregate market capitalization; average monthly turnover (annualized) and the illiquidity parameter λ (estimated over the last five years of our sample from 2013 to 2017). Panel B reports statistics for each of the 4 strategies we cover in this paper. λ has no unit. Volume, turnover, persistence and Sharpe ratios are annualized. The “raw” Sharpe ratio is that of the long-short portfolio where each stock is simply weighted by the signal (this is by construction a dollar-neutral portfolio). The Markowitz Sharpe is computed using the back-testing procedure described in Section 4.1.

Table 2: Capacity: Results from the Calibrated Model

		Mid					Large				
		Frictionless Sharpe	ϕ ($\times 10^{-3}$)	λ ($\times 10^{-6}$)	Capacity (\$ bn)	Capacity (Monop., \$ bn)	Frictionless Sharpe	ϕ ($\times 10^{-3}$)	λ ($\times 10^{-6}$)	Capacity (\$ bn)	Capacity (Monop., \$ bn)
Book-to-market											
	1990-2001	.43	.9	165	.33	.38	.14	.6	88	0	0
	2002-2017	-.19	.7	27	0	0	-.3	.5	5	0	0
Low vol											
	1990-2001	.41	1	165	.16	.19	.22	1.4	88	0	0
	2002-2017	1.1	1.7	27	14	19	.77	1.4	5	43	52
Momentum											
	1990-2001	.56	6.9	165	.027	.025	.35	7.1	88	.02	.016
	2002-2017	.46	6.8	27	.054	.064	.37	6.7	5	.064	.073
Cash-Flows											
	1990-2001	1.4	1.4	165	5.3	7.4	.51	.9	88	1.3	1.6
	2002-2017	.78	1.2	27	12	14	.52	.8	5	37	43

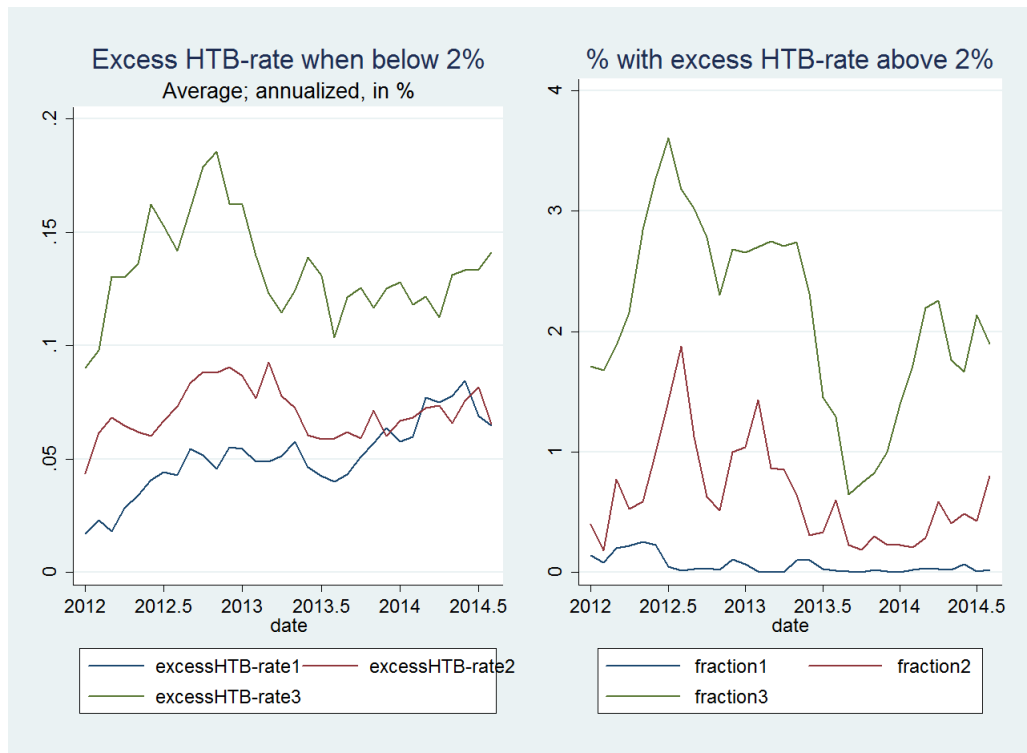
Note: This Table reports the theoretical capacity of each strategy by sub-period and by pool. We use the formula derived in the main text in Proposition 1 and define capacity as the annualized dollar volatility at which the realized Sharpe (SR) becomes equal to 0.3 For each sub-period, and each strategy, we compute the persistence parameter ϕ by regressing the signal on its lagged value, using monthly data. We compute the price impact coefficient λ by taking the median of $(2/15.9) \times \text{daily shares traded/daily vol}^2$ over all months-stocks in the pool and the period. λ is thus period and pool specific, but it the same across strategies. Finally, we estimate the Sharpe ratio of the strategy using the simplified Markowitz procedure described in Section 3.6: This procedure assumes a simplified correlation structure of returns, and no transaction costs. We use the estimated SR^* , ϕ and λ to compute the \$ volatility reached with a Sharpe $SR = .3$: this is what we call the capacity of the signal in each pool-period. The capacity is set to 0 when the pure Sharpe is already lower than .3. Columns 1-5 are for the pool of “mid caps” while columns 6-10 are for the pool of large caps. In columns 1 and 6, we report the value of the frictionless Sharpe; In columns 2 and 7, the annualized value of ϕ (the formula uses the daily version); In columns 3 and 8 we report the λ , in columns 4 and 9 the \$ volatility reached when $N = 1000$, i.e., the capacity in the competitive case, and in columns 5 and 10 the \$ volatility reached when $N = 1$, i.e., the capacity in the monopoly case. Sharpes, volatility and ϕ are all annualized.

Table 3: Expected vs. Simulated Performance-Capacity Frontier for non-AR(1) Signals

Real signal	\$Volatility	SR/SR^*	
		Expected	Realized
AR(6)	vol=5	0.78	0.72
	vol=10	0.69	0.66
	vol=15	0.62	0.63
AR(4)	vol=5	0.79	0.78
	vol=10	0.72	0.71
	vol=15	0.66	0.64
AR(2)	vol=5	0.83	0.74
	vol=10	0.75	0.66
	vol=15	0.71	0.59

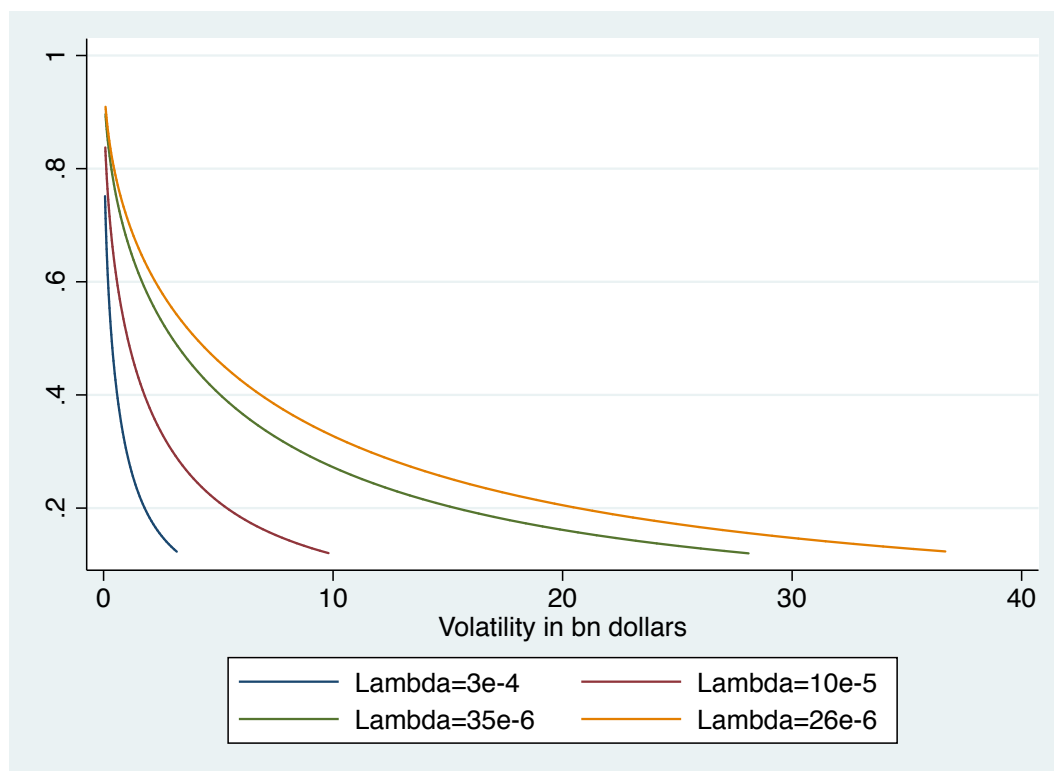
Note: This Table reports the Sharpe decay, $\frac{SR}{SR^*}$ as a function of dollar volatility (in annualized Bn dollar) for several synthetic signals that do not follow an AR(1) process. We perform monte-carlo simulation, where an AR(1) estimation is performed on the signal, and trading occurs according to those estimates. For each alternative process, we draw a large number of synthetic (signal, returns) datasets, on which we perform simulations. The number of stocks, liquidity, time length and volatility parameters are based on our mid-cap pool. For each draw of (signal, return), we estimate an AR(1) and simulate the performance of the trading rule recommended by the model. We vary γ to span a large spectrum of volatilities. The first three alternative signals correspond to an AR(L) process of the form: $s_{t+1} = \sum_{i=1}^L \frac{(1-\phi)}{L} s_{t-i} + \epsilon_{t+1}$. The ϕ parameter and signal noise are taken from the Cash-Flows signal.

Figure 1: Cost of shorting: Hard to borrow rates by size pools



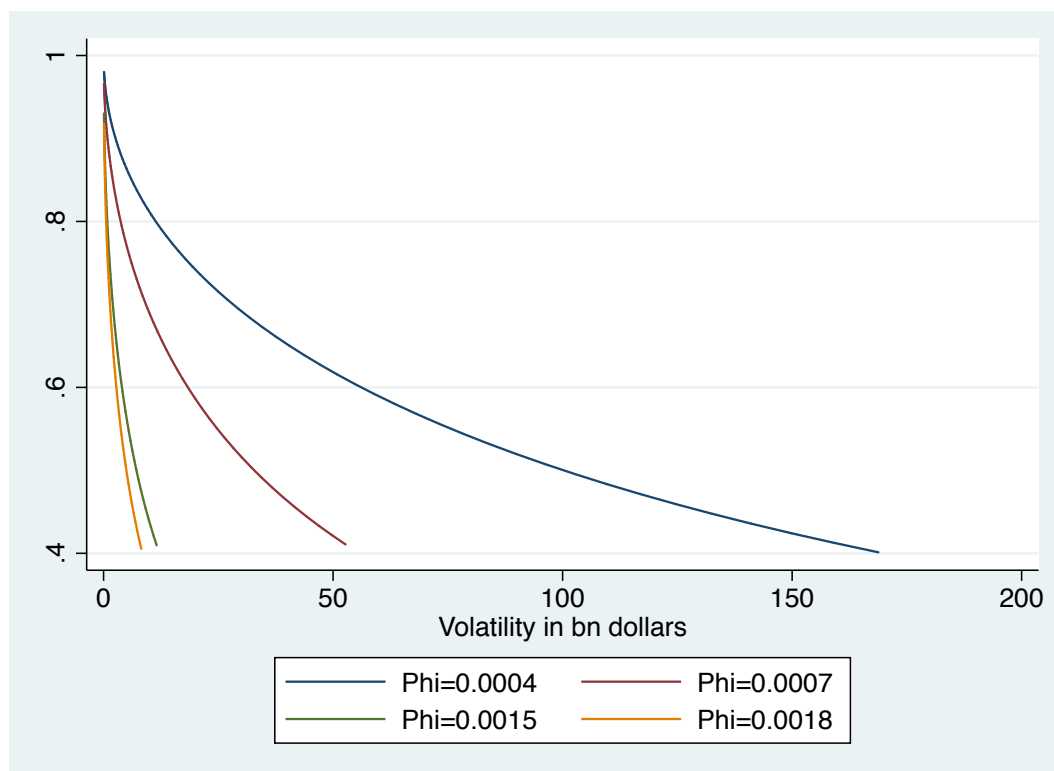
Note: Hard-to-borrow rates minus the General Collateral rate (or "easy-to-borrow rate") across size pools, using prop data. Rates are annualized, in %. The "General Collateral rate" corresponds to stocks which are not "hard-to-borrow" at a given point in time. We use the top 1500 US stocks sorted by size at each point in time. Pool 1 denotes the top 500 stocks, pool 2 is (500,1000) and pool 3 is (1000,1500).

Figure 2: Pool Liquidity and Capacity
Realized Sharpe ratio as a function of total \$ volatility



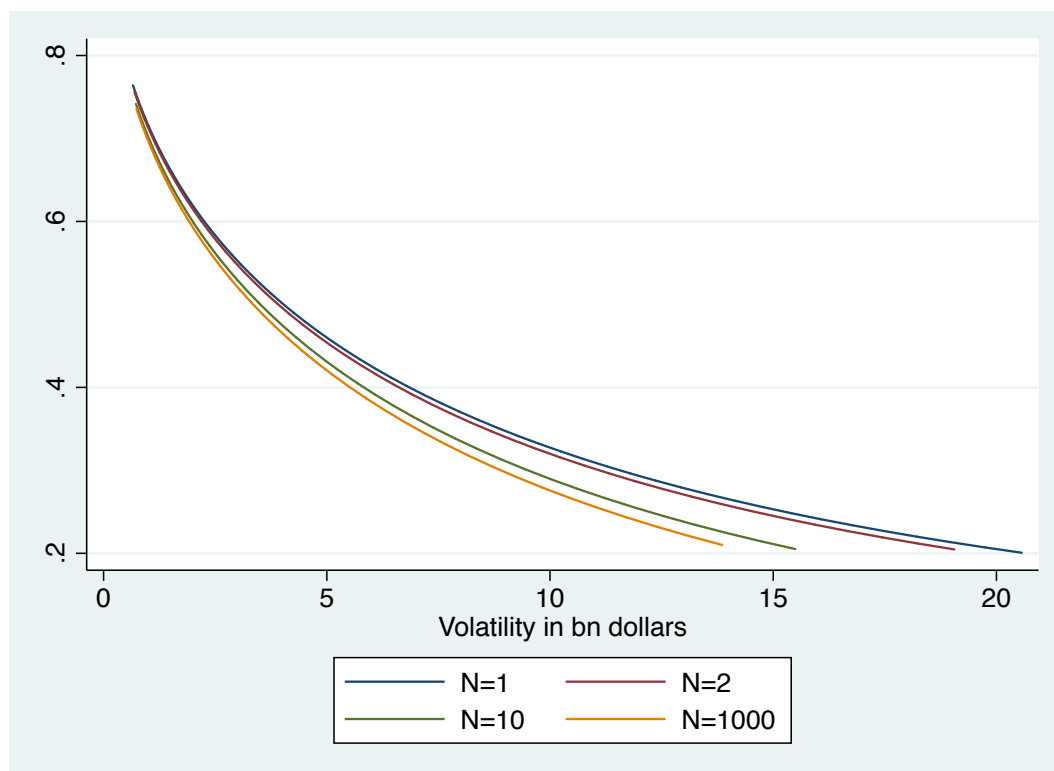
Note: This chart illustrates the impact of liquidity on strategy capacity. We consider a fictitious strategy whose frictionless Sharpe ratio is equal to 1, and whose persistence $\phi = 2.10^{-3}$ (25% in annualized terms), which roughly corresponds to the average persistence of our fundamental signals (see Table 1). Then, for each value of γ between 0 and 10^{-9} , we compute the Sharpe ratio and \$ volatility reached using the formula in Proposition 1 in the main text, considering the case without competition, i.e., $N = 1$. We use 4 different values of λ , which correspond roughly to the median λ in the mid pool in 1991-1995 (3 e-4), 1996-2000 (10 e-5), 2001-2005 (35 e-6) and 2013-2017 (26 e-6). We then draw the 4 frontiers with \$ volatility on the horizontal axis. Hence the blue line corresponds to the Sharpe-to-volatility frontier in the early 1990s, while the yellow line corresponds to the frontier in the later 2000s.

Figure 3: Signal Persistence and Capacity
Realized Sharpe ratio as a function of total \$ volatility



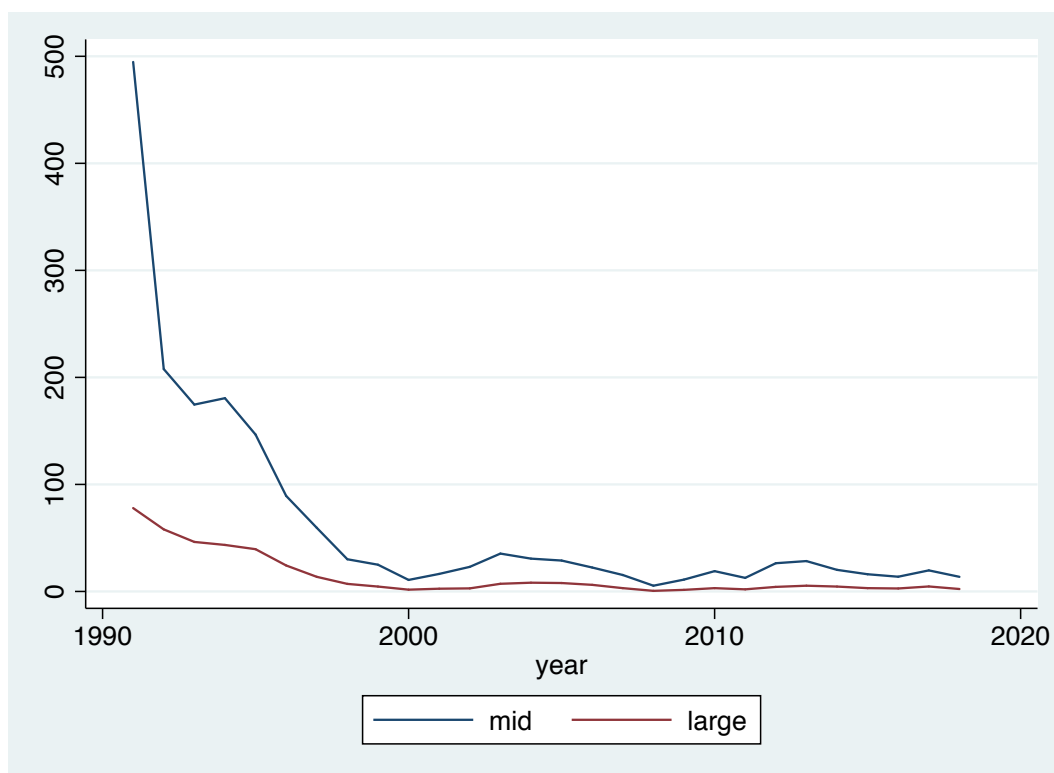
Note: This chart illustrates the impact of signal persistence on strategy capacity. We consider a fictitious strategy whose frictionless Sharpe ratio is equal to 1, and a pool whose price impact $\lambda = 26e - 6$, which corresponds to the liquidity estimated in the mid pool over the last five years of our sample 2013-2017 (see Table 1). Then, for each value of γ between 0 and 10^{-9} , we compute the Sharpe ratio and \$ volatility reached using the formula in Proposition 1 in the main text, considering the case without competition, i.e., $N = 1$. We use 4 different values of ϕ , which correspond to the ϕ in the 2000s for book-to-market in the large ($0.4 \cdot 10^{-3}$), book-to-market in the mid ($0.7 \cdot 10^{-3}$), low vol in the large ($1.5 \cdot 10^{-3}$), and low vol in the mid ($1.8 \cdot 10^{-3}$). We then draw the 4 frontiers with \$ volatility on the x axis.

Figure 4: Number of traders and Capacity
Realized Sharpe ratio as a function of total \$ volatility



Note: This chart illustrates the impact of the number of traders N on strategy capacity. We consider a fictitious strategy whose frictionless Sharpe ratio is equal to 1, whose price impact $\lambda = 26e - 6$, which corresponds to the liquidity estimated in the mid pool over the last five years of our sample 2013-2017, and whose persistence $\phi = 2.10^{-3}$ (25% in annualized terms), which roughly corresponds to the average persistence of our fundamental signals (see Table 1). Then, for each value of γ between 0 and 10^{-9} , we compute the Sharpe ratio and \$ volatility reached using the formula in Proposition 1 in the main text, considering different levels of N (the number of competing traders). We use 4 different values of N , which correspond to 1 (monopolistic case), 2, 10 and 1000 (virtually pure competitive). We then draw the 4 frontiers with \$ volatility on the x axis.

Figure 5: Illiquidity lambdas by size pool ($\times 10^{-6}$)



Note: Using CRSP daily data, we compute for each year a stock-level λ using the formula $\lambda_i = \frac{1}{8 \times volume_i \times \sigma_i^2}$. We divide stocks into two time-varying pools: the "large" pool is composed of the top 500 stocks by market capitalization and the "mid" pool is that of stocks ranked between 500 and 1500. For each pool of stocks, we then define for each year the pool's λ as the median of the liquidity parameters (λ_i) of stocks belonging to the pool. Scaling of lambdas is reported in ($\times 10^{-6}$).

Figure 6: Performance of Four Strategies: Large vs Midcaps

Panel A: Book-to-Market Big



Book-to-Market Mid



Panel B: Low Vol Big



Low Vol Mid



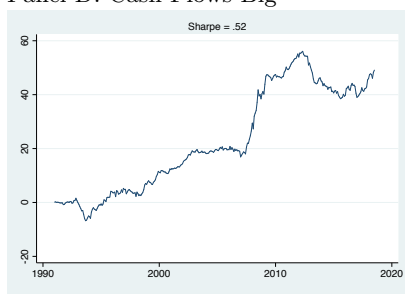
Panel C: Momentum Big



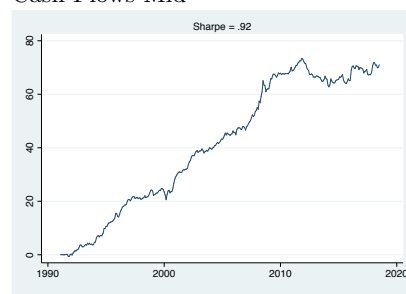
Momentum Mid



Panel D: Cash-Flows Big



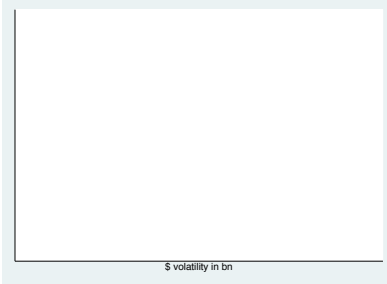
Cash-Flows Mid



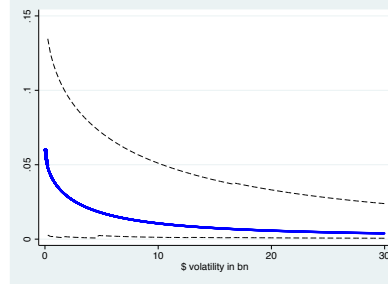
Note: These four panels correspond to four classical signals: Book-to-market, 1/rolling volatility, momentum and cash-flows. Portfolio weights are computed as the rank of each stock, normalized so as to lie between -.5 and .5. We then hedge the portfolio using the simplified Markowitz procedure described in Section 3.6: This procedure assumes a simplified correlation structure of returns, and no transaction costs.

Figure 7: Performance-Capacity Frontiers: Calibration Results

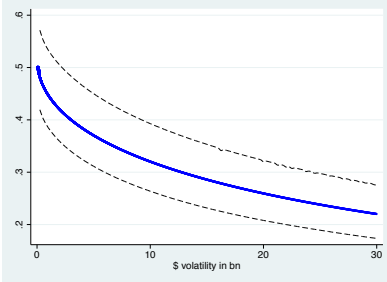
Panel A: Book-to-Market Big



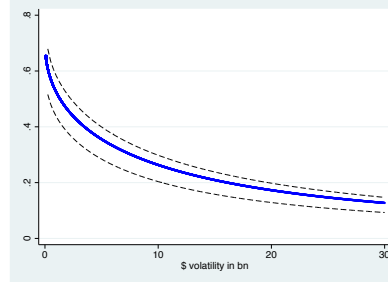
Book-to-Market Mid



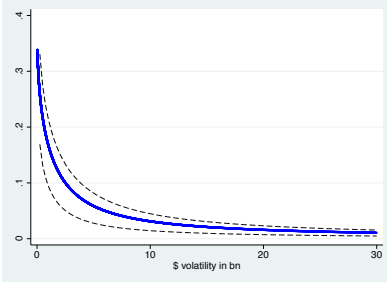
Panel B: Low Vol Big



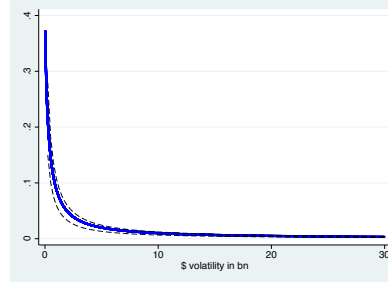
Low Vol Mid



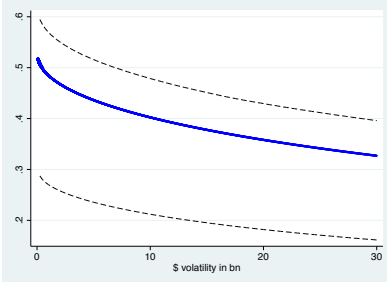
Panel C: Momentum Big



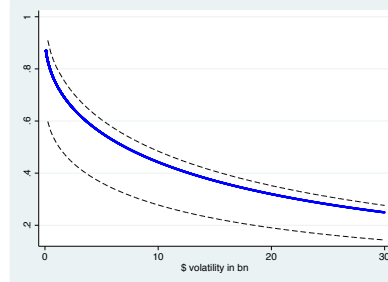
Momentum Mid



Panel D: Cash-Flows Big

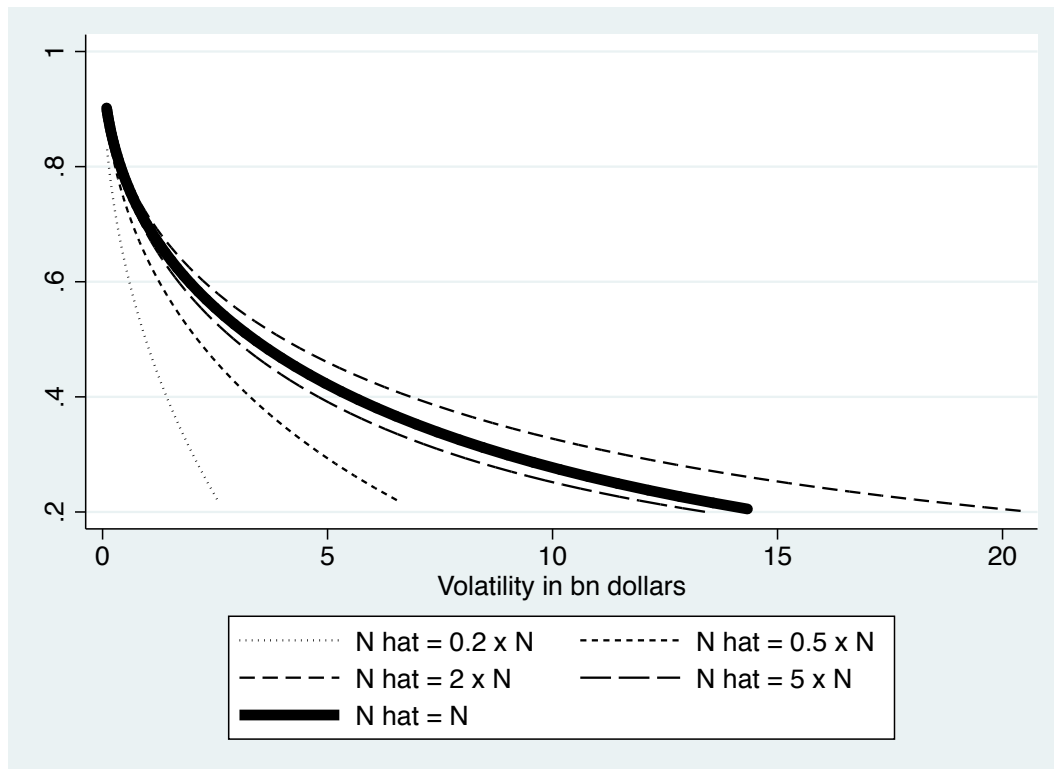


Cash-Flows Mid



Note: The thick line represents the Sharpe-to-volatility frontier obtained by estimating the model's parameters on the historical sample and plugging them in the formulae derived in Proposition 1. The dashed lines represent the top and bottom 5% Sharpe-to-volatility frontiers obtained in simulated data obtained by bootstrapping firms in each pool and calibrating the Sharpe-to-volatility frontier.

Figure 8: Effect of a misperception of the number of traders
Realized Sharpe ratio as a function of total \$ volatility



Note: In this Figure, we assume that $SR^* = 1$, $\phi = .002$ and $\lambda = 2.6 \cdot 10^{-5}$. To construct each one of the curves, we assume that the true number of traders $N = 100$, while traders believe that there are $\hat{N}$ traders. We explore 4 different values of $\hat{N} = .2N, .5N, 2N, 5N$ and compare them to the case where $\hat{N} = N = 100$ (the thick black curve).

APPENDIX

A Asymmetric case

In this appendix we consider the case where each trader's risk aversion coefficient is given by γ_k and can differ among traders. We start from the law of motion given by the FOC:

$$\gamma_k x_{k,t} = \gamma x_t^M - \frac{\lambda(1+\delta)}{2} (2\Delta x_{k,t} + \Delta x_{-k,t}) + \frac{\lambda}{2} (E_t \Delta x_{-k,t+1} + 2E_t \Delta x_{k,t+1})$$

where x_t^M is the aggregate Markowitz portfolio.

We postulate the following dynamics:

$$\begin{aligned} x_{k,t+1} &= (1 - \tau_k)x_{k,t} + \tau_k k_k x_{t+1}^M + \alpha_k x_{-k,t+1} + \beta_k x_{-k,t} \\ x_{-k,t+1} &= (1 - \tau_{-k})x_{-k,t} + \tau_{-k} k_{-k} x_{t+1}^M \end{aligned}$$

Substituting the above laws of motion into the FOC leads us to the values of the trading rule for $x_{k,t+1}$. In particular, it is clear that smaller traders (high γ_k will trade faster). However the problem is that the aggregate trading $x_t = x_{-k,t} + x_{k,t}$ is the sum of trading processes that respond to the same variations but at different speeds. It is not an AR(1) process, but instead the sum of AR(1)s with different half lives and common shocks (the Markowitz portfolio.). Therefore solving for the rational equilibrium for x_t appears challenging.

We propose an approach relying on a first order Taylor expansion to solve for the aggregate trading $x_t = \sum_k x_{k,t}$. Starting from the same FOC, we will assume in addition:

$$\begin{aligned} x_{k,t} &= \frac{x_t}{N} + \delta x_{k,t} \\ \gamma_k &= \frac{\sum_l \gamma_l}{N} + g_k = \gamma N + g_k \end{aligned}$$

where $\delta x_{k,t}$ and g_k are small. This ensures that:

$$\begin{aligned} \sum_k \delta x_{k,t} &= 0 \\ \sum_k g_k &= 0 \end{aligned}$$

so that we can neglect the second order terms and get:

$$\sum_k \gamma_k x_{k,t} \approx \gamma x_t$$

This allows us to collapse the law of motion (sum over k) as:

$$x_t = \underbrace{(\gamma \Sigma)^{-1} B s_t}_{x_t^M} - \frac{\lambda(1+\delta)}{2\gamma N} (N+1) \Delta x_t + \frac{\lambda}{2\gamma N} (N+1) E_t \Delta x_{t+1}$$

So, as in the text, x_t is like the monopoly case of Garleanu and Pedersen (2013) except with risk aversion multiplied by $\sqrt{\frac{N+1}{2N}}$.

Going back to the law of motion of trader k . Write that

$$\gamma_k x_{k,t} \approx \gamma x_t + \frac{x_t}{N} g_k + \gamma N \delta x_{k,t}$$

and inject the law of motion of x_t into the law of motion of $x_{k,t}$. We obtain:

$$\delta x_{k,t} = -\frac{g_k}{\gamma N} \frac{x_t}{N} - \frac{\lambda(1+\delta)}{2\gamma N} \Delta \delta x_{k,t} + \frac{\lambda}{2\gamma N} E_t \Delta \delta x_{k,t+1}$$

which has exactly the same structure as the aggregate trading rule, so that:

$$\delta x_{k,t} = (1 - \tau_n) \delta x_{k,t-1} - \tau_n k_n \frac{g_k}{\gamma N} \frac{x_t}{N}$$

with:

$$0 = \tau_n^2 + \tau_n \left(\delta + \frac{2\gamma N}{\lambda} \right) - \frac{2\gamma N}{\lambda}$$

$$k_n = \frac{\gamma N}{\gamma N + \frac{\lambda}{2} \tau_n \phi}$$

Therefore we get that the individual trading of trader k is given by:

$$x_{k,t} = \frac{x_t}{N} + \delta x_{k,t}.$$

The first component aims towards the Markowitz portfolio and trades at the speed of monopoly where γ is multiplied by $\frac{2N}{N+1}$, so faster as N goes up (as in the text). It also aims below the Markowitz portfolio but closer to it as N goes to infinity. The second component corrects for the fact that traders are smaller or bigger than average. It trades against aggregate trading, more aggressively so if the trader is smaller than average. This is to correct for the fact that a positive g_k trader does not want exactly the mean Markowitz but a slightly less levered portfolio. This second component also trades much faster than aggregate trading (with $2\gamma N$ instead of $\gamma(2N)/(N+1)$). It also trades closer to aim than aggregate trading.

This means that deviations from the homogeneous portfolio is traded faster than the homogeneous portfolio itself. However this additional speed of correction is the same for all traders, at the first order.

B Proof of Proposition 1

We note E is the unconditional expectation operator, which can be applied to any stationary process. We will use repeatedly two simple time-series properties: First if $E_t(y_{t+1}) = 0$ then for any stationary variable x_t , $E(x_t(y_{t+1})) = 0$; second, $E(x_t) = E(x_{t-1})$. Following Garleanu and Pedersen (2013) notation, we define $a = \tau\lambda$.

Step 1: Expected PNL.

We first compute the expected return of the optimal portfolio, without transaction costs:

$$\begin{aligned} ER_t &= E[x'_t r_{t+1}] \\ &= BE[x'_t s_t] \\ &= BE\left\{ \left[\left(1 - \frac{a}{\lambda}\right)x_{t-1} + \frac{a}{\lambda}x_t^{aim} \right]' [(1 - \phi)s_{t-1} + \epsilon_t] \right\} \\ &= \left(1 - \frac{a}{\lambda}\right)(1 - \phi)ER_{t-1} + \frac{a}{\lambda}BE(x_t^{aim'} s_t) \\ &= \left(1 - \frac{a}{\lambda}\right)(1 - \phi)ER_t + \frac{a}{\lambda}\theta B^2 E(s'_t \Sigma_u^{-1} s_t) \\ &= \frac{a/\lambda}{1 - (1 - \phi)(1 - a/\lambda)} \theta B^2 E(s'_t \Sigma_u^{-1} s_t) \end{aligned}$$

where

$$\theta = \frac{1}{\gamma + \frac{N+1}{2N}a\phi}$$

Step 2: Volatility of PNL.

Next, we compute the unconditional variance of R_t . Using the assumption $Es'_t s_t \ll \Sigma_u$, we get:

$$\begin{aligned} (Vol(R_t))^2 &= E[x'_t r_{t+1} r'_{t+1} x_t] \\ &= E[x'_t \Sigma_u x_t] \\ &= \left(1 - \frac{a}{\lambda}\right)^2 E(x'_{t-1} \Sigma_u x_{t-1}) + 2\frac{a}{\lambda}\left(1 - \frac{a}{\lambda}\right)E(x'_{t-1} \Sigma_u x_t^{aim'}) + \left(\frac{a}{\lambda}\right)^2 E(x_t^{aim'} \Sigma_u x_t^{aim}) \\ &= \frac{1}{1 - (1 - a/\lambda)^2} \left[2\frac{a}{\lambda}\left(1 - \frac{a}{\lambda}\right)E(x'_{t-1} \Sigma_u x_t^{aim'}) + \left(\frac{a}{\lambda}\right)^2 E(x_t^{aim'} \Sigma_u x_t^{aim}) \right] \end{aligned}$$

Now, we use $x_t^{aim} = \frac{1}{\gamma + \frac{N+1}{2N}a\phi} \Sigma_u^{-1} B s_t = \theta \Sigma_u^{-1} B s_t$:

$$\begin{aligned}
(Vol(R_t))^2 &= \frac{1}{1 - (1 - a/\lambda)^2} \left[2\frac{a}{\lambda}(1 - \frac{a}{\lambda})\theta BE(x'_{t-1}((1 - \phi)s_{t-1} + \epsilon_t)) + (\frac{a}{\lambda}\theta B)^2 E(s'_t \Sigma_u^{-1} s_t) \right] \\
&= \frac{1}{1 - (1 - a/\lambda)^2} \left[2\frac{a}{\lambda}(1 - \frac{a}{\lambda})\theta(1 - \phi)BE(x'_{t-1}(s_{t-1})) + (\frac{a}{\lambda}\theta B)^2 E(s'_t \Sigma_u^{-1} s_t) \right] \\
&= \frac{1}{(a/\lambda)(2 - a/\lambda)} \left[2\frac{a}{\lambda}(1 - \frac{a}{\lambda})\theta(1 - \phi) + (\frac{a}{\lambda}\theta)(1 - (1 - \phi)(1 - a/\lambda)) \right] ER_t \\
&= \frac{\theta}{(2 - a/\lambda)} \left[2(1 - \frac{a}{\lambda})(1 - \phi) + (1 - (1 - \phi)(1 - a/\lambda)) \right] ER_t \\
&= \frac{\theta}{(2 - a/\lambda)} [1 + (1 - \phi)(2(1 - a/\lambda) - (1 - a/\lambda))] ER_t \\
&= \frac{\theta}{(2 - a/\lambda)} [1 + (1 - \phi)(1 - a/\lambda)] ER_t
\end{aligned}$$

or:

$$Vol(R_t) = \frac{\theta[1 + (1 - \phi)(1 - a/\lambda)]}{2 - a/\lambda} \frac{E(R_t)}{Vol(R_t)}$$

Step 3: Expected transaction costs.

$$\begin{aligned}
E(TC_t) &= \frac{\lambda}{2} E \Delta x'_t \Sigma_u \Delta x_t \\
&= \lambda [E(x'_t \Sigma_u x_t) - E(x'_{t-1} \Sigma_u x_t)] \\
&= \lambda [E(x'_t \Sigma_u x_t) - E(x'_{t-1} \Sigma_u ((a/\lambda)x_t^{aim} + (1 - a/\lambda)x_{t-1}))] \\
&= \lambda(a/\lambda) [E(x'_t \Sigma_u x_t) - E(x'_{t-1} \Sigma_u x_t^{aim})] \\
&= a [E(x'_t \Sigma_u x_t) - \theta(1 - \phi)E(R_t)] \\
&= a [(Vol(R_t))^2 - \theta(1 - \phi)E(R_t)]
\end{aligned}$$

where we use:

$$E(x'_{t-1} \Sigma_u x_t^{aim}) = E(x'_{t-1} \Sigma_u \theta \Sigma_u^{-1} B s_t) = B \theta (1 - \phi) E(x'_{t-1} s_{t-1}) = \theta(1 - \phi) E R_{t-1} = \theta(1 - \phi) E R_t$$

This can be conveniently expressed by unit of volatility:

$$\begin{aligned}
\frac{E(TC_t)}{Vol(R_t)} &= a \left[(Vol(R_t)) - \theta(1 - \phi) \frac{E(R_t)}{Vol(R_t)} \right] \\
&= a \left[\frac{\theta[1 + (1 - \phi)(1 - a/\lambda)]}{2 - a/\lambda} - \theta(1 - \phi) \right] \frac{E(R_t)}{Vol(R_t)} \\
&= a\theta \left[\frac{1 + (1 - \phi)(1 - a/\lambda) - (1 - \phi)(2 - a/\lambda)}{2 - a/\lambda} \right] \frac{E(R_t)}{Vol(R_t)} \\
&= \frac{a\theta\phi}{2 - a/\lambda} \frac{E(R_t)}{Vol(R_t)}
\end{aligned}$$

Step 4: PNL Sharpe Ratio.

$$\begin{aligned}
SR &= \frac{E(R_t - TC_t)}{Vol(R_t)} \\
&= \left[1 - \frac{a\theta\phi}{2 - a/\lambda} \right] \frac{ER_t}{Vol(R_t)}
\end{aligned}$$

$$\begin{aligned}
\left(\frac{E(R_t)}{Vol(R_t)}\right)^2 &= \frac{2 - a/\lambda}{\theta[1 + (1 - \phi)(1 - a/\lambda)]} ER_t \\
&= \frac{2 - a/\lambda}{\theta[1 + (1 - \phi)(1 - a/\lambda)]} \frac{\theta a/\lambda}{1 - (1 - \phi)(1 - a/\lambda)} B^2 E(s'_t \Sigma_u^{-1} s_t) \\
&= \frac{2 - a/\lambda}{[1 + (1 - \phi)(1 - a/\lambda)]} \frac{a/\lambda}{1 - (1 - \phi)(1 - a/\lambda)} B^2 E(s'_t \Sigma_u^{-1} s_t) \\
&= \frac{1 - (1 - a/\lambda)^2}{1 - ((1 - \phi)(1 - a/\lambda))^2} B^2 E(s'_t \Sigma_u^{-1} s_t)
\end{aligned}$$

or

$$\frac{E(R_t)}{Vol(R_t)} = \left(\frac{1 - (1 - a/\lambda)^2}{1 - ((1 - \phi)(1 - a/\lambda))^2} \right)^{1/2} SR_t^*$$

The formula for the Sharpe Ratio follows directly.

The last step consists of computing the portfolio volatility:

$$\begin{aligned}
Vol(R_t) &= \theta \frac{(1 + (1 - \phi)(1 - a/\lambda))}{2 - a/\lambda} \frac{E(R_t)}{Vol(R_t)} \\
&= \frac{1}{\gamma + \frac{N+1}{2N} a \phi} \frac{1 + (1 - \phi)(1 - a/\lambda)}{2 - a/\lambda} \left(\frac{1 - (1 - a/\lambda)^2}{1 - ((1 - \phi)(1 - a/\lambda))^2} \right)^{1/2} SR^* \\
&= \frac{1}{\gamma + \frac{N+1}{2N} a \phi} \frac{1 + (1 - \phi)(1 - a/\lambda)}{2 - a/\lambda} \left(\frac{(2 - a/\lambda)(a/\lambda)}{(1 - (1 - \phi)(1 - a/\lambda))^2} \right)^{1/2} SR^* \\
&= \frac{1}{\gamma + \frac{N+1}{2N} a \phi} \left[\frac{a/\lambda}{2 - a/\lambda} \frac{1 + (1 - \phi)(1 - a/\lambda)}{1 - (1 - \phi)(1 - a/\lambda)} \right]^{1/2} SR^*
\end{aligned}$$

QED.

C Proof of Proposition 2

Large investment approximation

First, we only make the large investment approximation (and this let ϕ be large). We first compute the approximation for the trading rate $\tau = a/\lambda$. It is given by:

$$\begin{aligned}
\tau &= \frac{1}{2} \cdot \left(- \left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} + \delta \right) + \sqrt{\left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} + \delta \right)^2 + 4 \frac{\gamma}{\lambda} \frac{2N}{(N+1)}} \right) \\
&= \frac{1}{2} \cdot \left(- \left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} + \delta \right) + \sqrt{\left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} \right)^2 + 2 \frac{\gamma}{\lambda} \delta \frac{2N}{(N+1)} + \delta^2 + 4 \frac{\gamma}{\lambda} \frac{2N}{(N+1)}} \right) \\
&= \frac{1}{2} \cdot \left(- \left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} + \delta \right) + \sqrt{\left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} \right)^2 + 2 \frac{\gamma}{\lambda} \frac{2N}{(N+1)} (\delta + 2) + \delta^2} \right)
\end{aligned}$$

Under the large investment approximation, $\delta \ll (\gamma/\lambda)^{1/2} \ll 1$, so the trading rate simplifies into:

$$\tau \approx \frac{1}{2} \cdot \left(- \left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} + \delta \right) + \sqrt{\left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} \right)^2 + \delta^2 + 4 \frac{\gamma}{\lambda} \frac{2N}{(N+1)}} \right)$$

Also, $\frac{\gamma}{\lambda} \ll 1$ thus:

$$\tau \approx \frac{1}{2} \cdot \left(- \left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} + \delta \right) + \sqrt{\delta^2 + 4 \frac{\gamma}{\lambda} \frac{2N}{(N+1)}} \right)$$

Finally, $\frac{\gamma}{\lambda} \gg \delta^2$ which leads to the simple form:

$$\tau \approx \frac{1}{2} \cdot \left(- \left(\frac{\gamma}{\lambda} \frac{2N}{(N+1)} + \delta \right) + \sqrt{4 \frac{\gamma}{\lambda} \frac{2N}{(N+1)}} \right)$$

Thanks to the assumptions that $\frac{\gamma}{\lambda} \ll 1$ and $\frac{\gamma}{\lambda} \gg \delta^2$, the term in square root dominates so that:

$$\tau \approx \sqrt{\frac{\gamma}{\lambda} \frac{2N}{(N+1)}}$$

Given the above formula for the trading speed τ , the large investment approximation implies that: $\tau \ll 1$ and $\tau \gg \delta$. Only a few percent of the portfolio are traded every day, but this churning rate is much larger than the daily discount rate.

We then compute the Sharpe ratio. The first term of the Sharpe Ratio formula is given by:

$$1 - \frac{\tau\phi}{\frac{\gamma}{\lambda} + \tau\phi \frac{N+1}{2N}} \frac{1}{1 + (1 - \tau)} = 1 - \frac{\phi \frac{2N}{(N+1)}}{\phi + \tau} \frac{1}{2 - \tau} \approx \frac{\tau + \phi \left(1 - \frac{N}{(N+1)}\right)}{\phi + \tau}$$

since $\tau \ll 1$.

The second term of the Sharpe ratio is given by

$$\left[\frac{1 - (1 - \tau)^2}{1 - (1 - \tau)^2(1 - \phi)^2} \right]^{1/2} = \tau^{1/2} \left[\frac{2 - \tau}{1 - (1 - \tau)^2(1 - \phi)^2} \right]^{1/2} \approx \frac{\tau^{1/2}}{1 - \phi} \cdot \left[\frac{1}{\frac{(1 - (1 - \phi)^2)}{2(1 - \phi)^2} + \tau} \right]^{1/2}$$

where we also use the fact that $\tau \ll 1$.

Combining the two parts, we obtain that:

$$SR \approx \left(\frac{1}{\phi + \tau} \right) \frac{\left(\tau^{3/2} + \phi \tau^{1/2} \left(1 - \frac{N}{(N+1)}\right) \right)}{1 - \phi} \left(\frac{1}{\frac{(1 - (1 - \phi)^2)}{2(1 - \phi)^2} + \tau} \right)^{1/2} SR^*$$

which is the expression for the Sharpe ratio under the large investment approximation.

We now compute volatility:

$$Vol = \frac{1}{\lambda} \frac{\frac{2N}{(N+1)}}{\tau^2 + \tau\phi} \left[\frac{\tau}{2 - \tau} \frac{1 + (1 - \phi)(1 - \tau)}{1 - (1 - \phi)(1 - \tau)} \right]^{1/2} SR^*$$

Using the assumption that $\tau \ll 1$ we get that:

$$Vol \approx \frac{1}{\lambda \tau^{1/2}} \frac{\frac{2N}{(N+1)}}{\tau + \phi} \left[\frac{1 - \phi/2}{\phi + \tau(1 - \phi)} \right]^{1/2} SR^*$$

This gives \$ volatility under the large investment approximation.

Slow signal approximation

Our second step consists of letting $\phi \ll 1$, which leads to:

$$SR \approx \frac{\tau^{3/2} \left(\frac{N+1+\frac{\phi}{\tau}}{(N+1)} \right)}{(\phi + \tau)^{3/2}} SR^*$$

$$Vol \approx \frac{\left(\frac{2N}{(N+1)} \right)}{\lambda \tau^2} \left[\frac{\tau}{\tau + \phi} \right]^{3/2} SR^*$$

Letting $N \rightarrow \infty$, we can rewrite the Sharpe ratio:

$$SR \approx \frac{\tau^{3/2}}{(\phi + \tau)^{3/2}} SR^*$$

and the volatility:

$$Vol \approx \frac{2}{\lambda \tau^2} \left[\frac{\tau}{\tau + \phi} \right]^{3/2} SR^*$$

Combining the two equations, it is easy to see that: $Vol = SR \frac{2}{\lambda \tau^2}$ which leads to the following expression that combines τ with the SR and the $\$$ volatility:

$$\tau = \sqrt{\frac{2SR}{\lambda Vol}}$$

The validity conditions in the Proposition immediately follow from $\delta \ll \tau \ll 1$.

To obtain the second result, we just need to reverse the expression for the Sharpe ratio obtained previously:

$$\tau = \phi \frac{1}{(SR^*/SR)^{2/3} - 1}$$

We then combine the two definitions of τ and obtain the formula in the proposition.

To show the third bullet point in the proposition, we first note $F(x) = x((SR^*/x)^{2/3} - 1)^2$. It is easy to see that F is convex (it is the square of a convex function) and decreasing. Hence, its inverse is concave and decreasing. Let us call it G , then:

$$SR = G\left(\frac{\lambda \phi^2}{2} Vol\right)$$

thus:

$$\begin{aligned} \frac{\partial^2 SR}{\partial \lambda \partial \phi} &= (\phi \cdot G' \cdot Vol + \lambda \phi^3 \cdot G'' \cdot Vol) \\ &< 0 \end{aligned}$$

since $G'' < 0$ and $G' < 0$. QED

For the last bullet point of the proposition, we go back to expressions of TC and ER in the proof of Proposition 1 and combine them with the approximation:

$$\begin{aligned} \frac{TC}{ER} &= \frac{\tau \phi}{(2 - \tau)} \frac{1}{\left(\gamma/\lambda + \frac{N+1}{2N} \tau \phi\right)} \\ &\approx \frac{\tau \phi}{2 - \tau} \frac{\left(\frac{2N}{N+1}\right)}{\tau^2 + \tau \phi} \\ &\approx \frac{\phi}{\phi + \tau} \frac{N}{N + 1} \end{aligned}$$

which is valid even when the “slow signal” approximation is not valid.

Letting $N \rightarrow \infty$, we combine the above expression with the formula of SR at the beginning of this proof, and obtain the stated result. QED

D Proof of Proposition 3

Let us first show the first bullet point. The fact that the true transaction cost parameter is λ^* only changes the transaction costs:

$$\frac{TC}{Vol(R_t)} = \frac{2\lambda^* \tau \phi}{2 - \tau} \frac{1}{\lambda((\gamma/\lambda) + \phi \tau)} \frac{E(R_t)}{Vol(R_t)}$$

We inject this new formula into the definition of the net Sharpe ratio, and obtain:

$$\frac{ER - TC}{Vol(R_t)} = \left(1 - \frac{2\lambda^* \phi}{\lambda(2 - \tau)} \frac{\tau}{(\gamma/\lambda) + \phi \tau}\right) \left(\frac{1 - (1 - \tau)^2}{1 - ((1 - \phi)(1 - \tau))^2}\right)^{1/2} SR_t^*$$

We then make the large investment approximation, so that $\tau \approx \sqrt{\gamma/\lambda} \ll 1$. This leads to:

$$SR = \left(1 - \frac{\lambda^*}{\lambda} \frac{\phi}{\tau + \phi}\right) \left(\frac{2\tau}{1 - (1 - \phi)^2(1 - 2\tau)}\right)^{1/2} SR_t^*$$

Then, we make the slow signal approximation: $\phi \rightarrow 0$, so that:

$$SR = \left(1 - \frac{\lambda^*}{\lambda} \frac{\phi}{\tau + \phi}\right) \left(\frac{\tau}{\phi + \tau}\right)^{1/2} SR_t^*$$

The algebra for volatility follows similar steps.

The second result is shown using a similar method, except that, if the investor makes a mistake on ϕ , it does not affect his trading speed but only the aimed portfolio. QED.

E Proof of Proposition 4

Given the dynamics in equation (14), the formulae developed in Appendix B are still valid, except that we need to replace a/λ by π and θ by $1/\gamma$. The formula for the Sharpe ratio becomes:

$$\begin{aligned} SR(\gamma) &= \left[1 - \frac{\pi\lambda\phi}{\gamma} \frac{1}{1 + (1-\pi)}\right] \left[\frac{1 - (1-\pi)^2}{1 - (1-\pi)^2(1-\phi)^2}\right]^{1/2} SR^* \\ &\approx \left[\frac{\pi}{\pi + \phi}\right]^{1/2} SR^* \end{aligned}$$

while the formula for the volatility becomes:

$$\begin{aligned} Vol(\gamma) &= \frac{1}{\gamma} \left[\frac{\pi}{2 - \pi} \frac{1 + (1-\pi)(1-\phi)}{1 - (1-\pi)(1-\phi)}\right]^{1/2} SR^* \\ &\approx \frac{1}{\gamma} \left[\frac{\pi}{\pi + \phi}\right]^{1/2} SR^* \end{aligned}$$

We combine the two equations and obtain:

$$Vol = \frac{2N}{N+1} \frac{SR}{\lambda\phi} \left[\left(\frac{SR^*}{SR}\right)^2 - 1\right]$$

QED.

F A simple implementation of Markowitz hedging

The variance-covariance matrix of *percent* returns is:

$$\Sigma = \sigma_M^2 \beta\beta' + \sigma^2 Id$$

Lemma F.1. *The inverse of the variance covariance matrix of percent returns is:*

$$(\Sigma^{\%})^{-1} = \frac{1}{\sigma^2} \left(I - \frac{\sigma_M^2}{\sigma^2} \frac{\beta\beta'}{1 + \frac{\sigma_M^2}{\sigma^2} (\sum_i \beta_i^2)} \right)$$

and the dollar exposures of the Markowitz portfolio are:

$$P_t x_t^m = \frac{B^{\%}}{\gamma\sigma^2} \left\{ s_t - \frac{\sigma_M^2}{\sigma^2} \frac{\sum_i \beta_i s_i}{1 + \frac{\sigma_M^2}{\sigma^2} (\sum_i \beta_i^2)} \beta \right\}$$

Proof. Note that $(\beta\beta')^k = (\sum_i \beta_i^2)^{k-1} (\beta\beta')$ and

$$\Sigma^{-1} = \frac{1}{\sigma^2} \left(\sum_k \left(-\frac{\sigma_M^2}{\sigma^2}\right)^k (\beta\beta')^k \right)$$

□

It is then easy to compute the Sharpe ratio of the hedged portfolio:

$$(SR^*)^2 = B^2 s' \Sigma^{-1} s = \frac{B^2}{\sigma^2} \cdot \left(\sum_i s_i^2 - \sigma_M^2 \frac{\sum_i s_i \beta_i}{\sigma^2 + \sigma_M^2 \sum_i \beta_i^2} \right)$$

which is the formula we show in the text. QED